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RESEARCH ARTICLE

Confidence-Aware Test-Time Adaptation for Fault-Resilient Sound Source Localization Guided by Lightweight Diagnosis

INKYU AN
Department of Software, Kookmin University, Seoul 02707, South Korea
e-mail: inkyu.an@kookmin.ac.kr

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ABSTRACT Deep learning-based sound source localization systems typically rely on specific microphone array configurations, making them highly vulnerable to hardware failures. Unexpected sensor faults cause severe structural signal degradation, leading to substantial deterioration in localization accuracy. To address this challenge, we propose a robust Fault-Resilient Sound Source Localization framework that autonomously diagnoses sensor faults and adapts the model at test time without requiring ground-truth labels. The proposed system integrates two core modules: 1) a Lightweight Fault Diagnosis module, which employs a computationally efficient Frequency-Time Projection Network with a Fault Queue mechanism, and 2) a Diagnosis-Guided Confidence-Aware Test-Time Adaptation strategy, which leverages the highly reliable diagnosis results to apply interventions and utilizes a confidence thresholding mechanism to generate refined hard pseudo-labels for stable self-training. As a result, the proposed system reduces the mean absolute error, averaged across diverse models and fault conditions, from 21.99° to 9.91°, achieving a significant error reduction of 54.92%, thereby ensuring reliable robotic perception even in the presence of sudden hardware malfunctions. The proposed framework is further validated in a real-environment experiment using a physical microphone array with an authentic hardware fault, confirming its generalizability beyond synthetic fault conditions.

INDEX TERMS Acoustic signal processing, autonomous robots, deep learning, fault diagnosis, machine listening, microphone arrays.

I. INTRODUCTION

Sound source localization (SSL) is a fundamental capability for robotic perception and human-computer interaction, enabling embedded systems such as service robots and smart speakers to detect acoustic events and interact with users implicitly [1], [2], [3], [4], [5], [6]. Encompassing diverse tasks ranging from direction-of-arrival (DOA) estimation to environment-aware localization and acoustic mapping, these systems typically rely on multi-channel audio captured by microphone arrays, where spatial information is encoded in the time and phase differences between microphones.

Historically, it is well established that the accuracy of SSL is closely tied to the number of microphones [7], [8]; increasing the number of sensors generally improves spatial resolution but imposes a significant burden on computational cost and memory usage. For instance, high-end commercial arrays like the 64-channel Eigenmike can achieve precise localization but require processing power often unavailable on resource-constrained embedded platforms [9]. To address this trade-off, recent advancements in deep learning have enabled robust SSL performance using compact array configurations [1], [2], [3]. Thanks to these data-driven approaches, it is now possible to achieve high-precision 2D DOA estimation (e.g., mean absolute error (MAE) of DOA of 2.65° [3]) using as few as four microphones.

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However, this reliance on minimal sensor configurations introduces a critical vulnerability: the system becomes highly sensitive to hardware failures. In real-world deployment scenarios, microphones are prone to various physical and electrical faults caused by hardware damage (e.g., broken wires), connection instability, or aging of vibrating membranes [10]. As detailed in Sec. III-B, these hardware issues manifest as distinct fault patterns, including *No Sound*, *Attenuation*, *Impulsive Noise*, *Continuous Noise*, and *Signal Replication*. The impact of such failures is substantial for deep learning-based models trained on healthy data. As demonstrated in our empirical analysis in Sec. VI-B2, a single microphone failure in a 4-channel array causes the MAE of DOA estimations, averaged across diverse models and fault conditions, to surge by approximately 238% (from 6.50° to 21.99°), rendering the localization system unreliable.

Even in the event of sudden hardware failure, autonomous systems must maintain an acceptable level of stability and functionality until repair or replacement can be performed. If a robot malfunctions due to an unexpected sensor fault, it could lead to erroneous decision-making, potentially resulting in safety accidents involving humans or damage to the environment. Consequently, extensive research has been conducted on fault detection and diagnosis (FDD) to ensure the reliability of various sensors, such as LiDAR [11], cameras [12], and IMUs [13], utilizing deep learning to handle complex signal patterns. However, research specifically addressing microphone array faults remains limited. While methods for diagnosing microphone faults exist, they often rely on traditional signal processing features, such as voice activity detection (VAD) inter-correlation [10]. These conventional approaches struggle to detect phase-related faults, such as channel copying (signal replication), because the replicated signals retain valid speech characteristics and energy levels.

Even if the microphone fault is accurately detected and diagnosed, a critical challenge remains in model recovery. Fine-tuning deep learning-based SSL models on corrupted multi-channel audio involves a significant domain shift, which typically requires access to ground-truth labels from the target domain. To overcome this limitation, extensive research has been conducted on test-time adaptation (TTA), which enables models to adapt to distribution shifts using only unlabeled test data [14], [15], [16], [17], [18], [19], [20], [21].

However, directly leveraging these existing TTA frameworks for SSL presents a non-trivial challenge. Most TTA methods are tailored for classification tasks, relying on entropy minimization or confidence maximization over discrete label spaces, which do not straightforwardly translate to the continuous regression nature of DOA estimation. Although recent studies have extended TTA to structured regression problems in computer vision—such as depth completion [22], optical flow [23], and pose estimation [24], [25]—these approaches predominantly exploit visual geometric constraints. Consequently, since these methods are inherently designed for the visual domain, they are not directly applicable to audio-based SSL tasks that require handling unique spectral and spatial degradations, particularly when hardware failures disrupt the inter-channel phase relationships.

To this end, we propose a robust **Fault-Resilient SSL framework** that can autonomously detect the specific fault type and adapt the pre-trained model to the target domain without external intervention. Previous research on microphone fault diagnosis has primarily focused on specific features, while existing TTA approaches are ill-suited for SSL under hardware faults as they do not account for the structural signal degradation. This paper presents a tightly coupled fault-resilient SSL framework in which fault semantics explicitly guide the adaptation process.

The main contributions of this paper are summarized as follows:

- **Lightweight and Reliable Fault Diagnosis (Sec. IV-A):** We propose a computationally efficient Frequency-Time Projection Network integrated with a robust Fault Queue mechanism to ensure precise fault identification.
- **Diagnosis-Guided Confidence-Aware TTA (Sec. IV-B):** We propose a confidence-aware TTA framework based on hard pseudo-labeling, which enables stable self-training for SSL under microphone faults. Furthermore, we introduce a diagnosis-guided adaptation strategy that leverages fault semantics to enhance robustness against severe structural degradations.

We rigorously validate both the fault diagnosis and adaptation modules with synthetically injected faults (Sec. VI). First, we verify that the proposed Lightweight Fault Diagnosis model can be stably trained to yield precise fault classification in Sec. VI-A. Subsequently, through extensive experiments in Sec. VI-B, we demonstrate the superiority of our TTA method over the baselines; with all proposed components enabled, the system achieves a significant error reduction of 54.92% (MAE improving from 21.99° to 9.91°), averaged across diverse models and fault conditions.

We further confirm the practical stability of our system for real-time embedded deployment through sensitivity analyses covering threshold initialization, dynamic thresholding, silence threshold, batch size, and threshold sensitivity (from Sec. VI-B3 to Sec. VI-B7), and analyze the system's robustness under misdiagnosis scenarios in Sec. VI-B8.

Beyond single-channel failure conditions, we extend our evaluation to simultaneous dual microphone failure scenarios in Sec. VI-C, identifying the boundary between recoverable and irrecoverable failure states imposed by the geometric constraint requiring at least three operational microphones for reliable 2D DOA estimation. Finally, we validate the generalizability of the framework beyond the SSLR dataset through a real-environment experiment using a physical microphone array with an authentic hardware fault in Sec. VI-D.

II. RELATED WORKS

A. SOUND SOURCE LOCALIZATION

Research in SSL can be broadly categorized into DOA estimation, environment-aware localization, and mobile robot-based acoustic mapping.

DOA estimation has been extensively studied as a core problem in SSL, and recent advances in deep learning have significantly accelerated progress in this direction. By learning the mapping between acoustic features and source directions, deep learning-based approaches have demonstrated improved robustness to noise and reverberation, while enabling accurate localization even with a small number of microphones, such as compact and minimally redundant microphone arrays. He et al. [1] employed deep neural networks for multiple speaker detection and localization, demonstrating robust performance in noisy conditions. Wang et al. [2] introduced a mechanism to apply attention to speech-dominant frequency bins in generalized cross correlation-phase transform (GCC-PHAT) features, enhancing localization accuracy. Furthermore, An et al. [3] proposed a microphone pair training strategy to achieve robust SSL across diverse array configurations, reducing the dependency on specific array geometries during training.

Beyond DOA estimation, researchers have leveraged environmental cues and auxiliary sensor data to handle complex acoustic scenes. An et al. [4], [5] incorporated diffraction and reflection models into the localization framework, utilizing back-propagation signals and environmental geometry to localize sources even in non-line-of-sight conditions. In a data-driven context, Liu et al. [6] proposed a self-supervised incremental learning framework to adaptively learn the acoustic-to-location mapping in complex indoor environments, reducing the reliance on extensive annotated data.

For mobile robots, integrating spatio-temporal information over multiple frames is crucial for consistent tracking. Su et al. [26] proposed a real-time 3D sound source mapping framework using linear arrays, accumulating DOA estimates while the robot moves. More recently, Fu et al. [27] introduced I-ASM, an iterative acoustic scene mapping approach that enhances auditory perception in complex indoor environments by continuously refining the acoustic map.

Despite these advancements, a fundamental assumption across these methods is the structural integrity of the microphone array. Consequently, any deviation from the expected configuration, such as hardware failures (e.g., dead channels, gain mismatch), disrupts the spatial correlation structure, leading to severe degradation in localization accuracy. This vulnerability has received limited attention in existing SSL literature, yet poses a significant hurdle for long-term robot operation. We experimentally quantify this performance drop caused by sensor faults in Sec. VI-B.

B. SENSOR FAULT DETECTION AND DIAGNOSIS

Hardware devices operating in real-world environments are prone to unexpected failures. Consequently, substantial research has been conducted on fault detection and diagnosis (FDD) to ensure system reliability and safety [28], [29]. With the recent advancements in artificial intelligence, deep learning-based FDD has been actively investigated to handle complex signal patterns.

For instance, Zhao et al. [30] proposed a method using strong residual convolutional neural networks (CNNs) with adaptive thresholds to reliably diagnose faults in rotating machinery (e.g., motors) from vibration signals, even under noisy conditions. Similarly, Mezair et al. [31] introduced a hybrid model combining CNN, long short-term memory (LSTM), and graph neural network (GNN) to detect anomalies in industrial processes using heterogeneous time-series sensor data. Ali et al. [32] utilized short-time Fourier transform (STFT) and ensemble deep learning for motor fault diagnosis. In the acoustic domain, Ha et al. [33] employed a microphone array and a thermal imaging camera to detect faults in transmission lines. While these studies demonstrate the efficacy of deep learning in industrial diagnosis, their primary objective is to identify anomalies in the surrounding environment (e.g., motor malfunction, transmission line defects). Consequently, they typically operate under the assumption that the sensors themselves are functioning correctly, leaving the issue of internal sensor failures unaddressed.

Distinct from fault diagnosis targeting external environmental phenomena, a separate line of research has focused on ensuring the integrity of the sensing hardware itself. In visual perception, research has been conducted to detect and handle lens soiling in LiDAR [11] and camera sensors [12]. For satellite attitude determination, Chen et al. [13] proposed a recovery method fusing Kalman filters and neural networks to handle IMU sensor faults. In multi-modal sensor fusion, Zhang et al. [34] demonstrated robust performance maintenance even when a specific sensor (e.g., camera) fails or is missing. Cicero et al. [35] addressed anomaly detection in label-scarce IoT sensor networks using a U-Net architecture and sliding window-based online learning. Furthermore, in the context of antenna arrays, which share similar array signal processing principles with microphone arrays, Zheng et al. [36] utilized a ResNet architecture to robustly detect faults in noisy environments.

Despite the critical impact of array integrity on SSL performance, the majority of existing FDD research using microphone arrays remains focused on localizing the source of mechanical faults rather than diagnosing the microphone array itself [33], [37], [38], [39]. While methods for diagnosing microphone faults exist, they often rely on traditional signal processing features, such as VAD inter-correlation, which are insufficient for detecting phase-related faults like channel copying, where replicated signals preserve valid spectral and energy characteristics [10]. Recently, Zhou et al. [40] addressed DOA estimation under array imperfections via supervised transfer learning. Although this approach attempts to adapt to the target domain, it does not address automatic fault detection and requires labeled

data from the target domain for fine-tuning, which limits its applicability in autonomous deployment settings. Therefore, a unified framework that autonomously detects sensor faults and enables unsupervised model adaptation is required.

C. TEST-TIME ADAPTATION

Since ground-truth labels are unavailable for corrupted target-domain data during deployment, adaptation must be performed in an unsupervised manner. Test-time adaptation (TTA) has emerged as a promising paradigm to address this by adapting models to shifted target domains without access to the target labels. Traditionally, the majority of TTA methodologies have been developed primarily for classification tasks. Early approaches focused on entropy minimization to reduce uncertainty in discrete prediction distributions, as demonstrated by TENT [14] and MEMO [15]. Others addressed catastrophic forgetting by selecting reliable low-entropy samples [16] or mitigating sharp loss minima [17]. In parallel, pseudo-labeling strategies have been actively explored for classification, utilizing techniques like nearest neighbor information [18], teacher-student architectures (CoTTA) [19], and confidence-based filtering [21] to guide self-training.

Recognizing the need for adaptation in continuous output spaces, recent efforts have extended TTA to regression problems, particularly within the computer vision domain. Unlike classification, these tasks require estimating continuous values where standard entropy minimization is not directly applicable. Representative examples include depth completion [22], optical flow estimation [23], and pose estimation [24], [25]. These methods typically leverage structural constraints or auxiliary guidance specific to visual modality (e.g., geometric consistency) to drive the adaptation process.

However, directly applying these existing TTA frameworks to SSL remains non-trivial due to inherent task and domain discrepancies. First, SSL is fundamentally a regression problem that estimates continuous source locations (e.g., DOA). This precludes the direct applicability of prevalent classification-based TTA methods, as their core mechanisms—such as minimizing the entropy of discrete class probabilities—do not straightforwardly translate to continuous regression outputs. Second, adaptation strategies tailored for vision-based regression do not readily generalize to audio-based SSL, especially under microphone array failures. The spatial cues in audio signals (e.g., phase differences) are fundamentally different from visual features. Consequently, the direct transfer of methods designed for the visual modality to the SSL problem is constrained by these inherent domain discrepancies.

It is also important to distinguish TTA from fault-augmented retraining, an alternative strategy that addresses domain shifts by augmenting the training data with fault-injected samples and retraining the model offline. While fault-augmented retraining can improve robustness to anticipated fault conditions, it requires access to ground-truth labels from the target domain and must be performed prior to deployment, making it fundamentally unsuitable for autonomous systems that must respond to sudden hardware failures in real time without human intervention. In contrast, TTA adapts the pre-trained model online at test time using only unlabeled streaming data, without requiring any prior knowledge of the fault type or access to labeled data from the target domain. This distinction makes TTA a uniquely appropriate paradigm for the fault-resilient SSL problem addressed in this work.

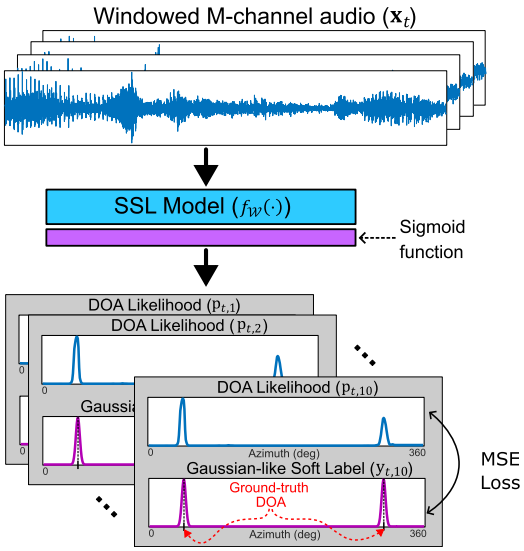


FIGURE 1. Overview of the deep learning-based sound source localization (SSL) framework. The model processes windowed *M*-channel audio signals at the current frame *t* to estimate a sequence of direction-of-arrival (DOA) likelihoods for *Q* consecutive intra-frame steps (e.g., $Q = 10$). Gaussian-like soft label sequences are employed to represent the ground-truth trajectory over time. The network utilizes a sigmoid activation function and is optimized by minimizing the sequence-wise mean squared error (MSE) loss between the predicted likelihood sequence and the Gaussian target sequence.

III. PROBLEM FORMULATION

A. SOUND SOURCE LOCALIZATION TASK

We consider a robot equipped with a microphone array consisting of M microphones. Let $s(n)$ denote a sound source signal in the time domain. Under the far-field assumption, the signal received by the m-th microphone, $x_m(n)$, can be modeled as:

$$x_m(n) = s(n - \tau_m(\theta)) + v_m(n), \tag{1}$$

where $\tau_m(\theta)$ is the time delay of arrival relative to the source direction θ and the m-th microphone position, and $v_m(n)$ denotes additive background noise. While the general DOA θ encompasses both azimuth and elevation in 3D space, we simplify the problem to 2D planar localization by assuming the source and the robot lie on the same horizontal plane. Thus, in this work, θ represents the azimuth angle solely.

Following recent data-driven approaches [1], [2], [3], we formulate the DOA estimation task as a supervised

learning problem for 2D planar localization. Crucially, to ensure stable estimation performance without front-back ambiguity, we assume that the microphone array consists of at least three microphones ($M \geq 3$). The continuous multi-channel signals are segmented into overlapping frames using a sliding window. Let t denote the time frame index. The goal is to learn a non-linear mapping function $f_{\mathcal{W}}(\cdot)$ parameterized by $\mathcal{W}$, which estimates a fine-grained sequence of DOAs within the observed signal frame. Specifically, we formulate the problem by dividing a single input frame t into Q consecutive intra-frame steps, allowing the model to capture the detailed trajectory within the frame duration.

Specifically, for the t-th time frame, the system takes the windowed multi-channel audio signals $\mathbf{x}_t \in \mathbb{R}^{M\times N}$ as input, where N denotes the number of samples in a frame. From this raw input, acoustic features $\mathbf{X}_t$ (e.g., STFT, inter-channel phase differences (IPD), or GCC-PHAT [41]) are computed and fed into the network. Note that the dimensionality of $\mathbf{X}_t$ depends on the feature extraction method; for instance, STFT-based features typically take the form $\mathbf{X}_t \in \mathbb{R}^{C\times L\times F}$, where L and F denote the number of time and frequency bins, respectively. Here, C represents the total number of feature channels, which varies depending on the specific model architecture and the combination of input features employed.

As illustrated in Fig. 1, the model outputs a sequence of spatial likelihood vectors $\mathbf{p}_{t,q} \in \mathbb{R}^K$ for each intra-frame step $q = 1, \ldots, Q$, defined over K discretized angular bins (e.g., $K = 360$ for 1° resolution). A pivotal design choice in our framework is the strategic adoption of this probabilistic representation over direct coordinate regression [42]. We identify that standard regression models, which output a single coordinate value, inherently lack the explicit uncertainty measures required for unsupervised adaptation. By leveraging this classification-like spatial likelihood estimation, we bridge this gap, utilizing the probabilistic output to compute confidence metrics (e.g., entropy) and generate pseudo-labels. This architectural decision serves as a fundamental enabler for applying confidence-driven TTA strategies to the SSL domain, as detailed in Sec. IV-B.

Consistent with prior studies [1], [2], [3], during the supervised pre-training phase, we employ a Gaussian-like soft labeling scheme. Accordingly, we define a sequence of ground-truth target vectors $\mathbf{y}_{t,q}$ corresponding to the q-th sub-step within the current frame t. Note that distinct hard pseudo-labels are employed later for the TTA process, as detailed in Sec. IV-B. To accommodate scenarios with multiple sound sources, let $\Theta_{t,q} = \{\theta^{gt}_{t,q,1}, \ldots, \theta^{gt}_{t,q,J}\}$ denote the set of ground-truth DOAs at the q-th step of frame t, where J represents the number of active sources. The target value for the k-th angular bin ϕ_k is defined by aggregating the Gaussian kernels for all active sources:

$$\mathbf{y}_{t,q,k} = \max_{\theta\in\Theta_{t,q}} \left(\exp\left(-\frac{d(\phi_k,\theta)^2}{2\sigma^2}\right)\right), \tag{2}$$

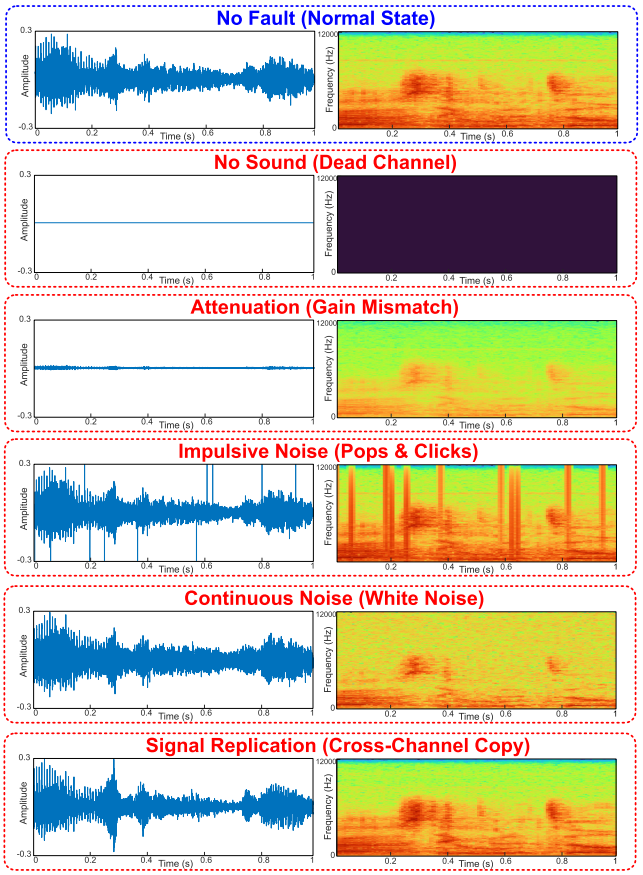


FIGURE 2. Visualization of microphone signals under normal and faulty conditions. The left column displays time-domain waveforms, and the right column shows the corresponding spectrograms. The color scale represents log-magnitude, ranging from dark blue (≈ -100 dB) to dark red (≈ 20 dB). While faults like *No Sound*, *Impulsive Noise*, and *Continuous Noise* exhibit distinct visual or spectral patterns, note that the *Signal Replication* fault (bottom) maintains a valid speech-like spectro-temporal structure similar to the *Normal State*. This visual similarity highlights the challenge of detecting replicated signals using conventional magnitude-based metrics.

where $d(\cdot)$ computes the angular distance, σ controls the spread of the label, and the max operation effectively overlays the spatial distributions of multiple sources.

The final layer of the network utilizes a Sigmoid activation function to ensure the output values lie within [0, 1]. To align with the training objectives of the baseline methods [1], [2], [3], the model is trained by minimizing the Mean Squared Error (MSE) loss between the predicted likelihood $\mathbf{p}_{t,q}$ and the Gaussian target $\mathbf{y}_{t,q}$ for each angle bin:

$$\mathcal{L}_{SSL} = \frac{1}{QK}\sum_{q=1}^{Q}\sum_{k=1}^{K}(\mathbf{y}_{t,q,k} - \mathbf{p}_{t,q,k})^2. \tag{3}$$

During the inference phase, the final DOA estimate is determined by identifying the angular bin corresponding to the global peak (or local peaks in multi-source scenarios) of the predicted likelihood vector $\mathbf{p}_{t,q}$. This formulation allows the model to robustly estimate the DOA for each time frame t even in the presence of noise and reverberation,

provided that the microphone array configuration remains consistent with the training phase.

B. MICROPHONE FAULT TYPES

In practical deployment scenarios, microphone arrays are susceptible to various hardware failures caused by physical damage, circuit degradation, or environmental factors [10]. To systematically analyze the impact of these failures, we model the faulty signal $\tilde{x}_m(n)$ received by the m-th microphone as a corrupted version of the clean signal $x_m(n)$:

$$\tilde{x}_m(n) = \alpha_m(n) \cdot x_m(n) + \beta_m(n), \tag{4}$$

where $\alpha_m(n)$ represents a (possibly time-varying) gain scaling factor (where $\alpha_m(n) = 1$ denotes a healthy state) and $\beta_m(n)$ represents additive fault noise.

Based on a prior study [10] and common hardware failure modes, we categorize five representative fault types commonly observed in practice, alongside the normal state (No Fault), as visualized in Fig. 2.

- **No Sound (Dead Channel):** This fault occurs due to disconnected cables, broken wires, or complete sensor failure. Mathematically, this corresponds to a complete loss of signal information, modeled as $\alpha_m(n) \approx 0$ and $\beta_m(n) \approx 0$. For a neural network, this results in a "dead" input channel containing only zeros or negligible thermal noise.
- **Attenuation (Gain Mismatch):** Microphone sensitivity can degrade over time due to dust accumulation on the acoustic port or aging of the vibrating membrane (diaphragm). This results in a consistent reduction in signal amplitude, modeled as $0 < \alpha_m(n) < 1$ with $\beta_m(n) = 0$.
- **Impulsive Noise (Pops & Clicks):** Caused by loose connections, electrical discharges, or mechanical impacts, this fault introduces sudden, high-energy spikes into the signal. Here, $\beta_m(n)$ appears as intermittent impulsive noise with high amplitude, temporarily overwhelming the source signal $x_m(n)$.
- **Continuous Noise (White Noise):** Degradation of internal amplifier circuits or electromagnetic interference can inject persistent wide-band noise. In this case, $\beta_m(n)$ is modeled as Gaussian white noise $\mathcal{N}(0, \sigma^2_{fault})$ with a variance significantly higher than the ambient noise floor, severely degrading the signal-to-noise ratio (SNR).
- **Signal Replication (Cross-Channel Copy):** This fault occurs when the signal from one microphone (e.g., m-th mic) is erroneously duplicated to another channel (e.g., j-th mic) due to data acquisition software errors or short circuits in the signal path, i.e., $\tilde{x}_j(n) = x_m(n)$, which cannot be captured by simple gain or additive noise models. Crucially, this fault type is difficult to detect using traditional methods. Previous works relying on magnitude or VAD features [10] often fail to identify this fault because the replicated signal possesses valid speech characteristics and energy levels (Sec. VI-A1). However, for SSL, this fault is particularly detrimental, as it destroys the unique phase and time difference of arrival (TDOA) information required for DOA estimation.

The occurrence of any of these faults fundamentally alters the input feature distribution $\mathbf{X}_t$. Consequently, the pre-trained mapping function $f_{\mathcal{W}}(\cdot)$, which assumes a healthy array configuration, fails to extract correct spatial cues, leading to a severe degradation in DOA estimation accuracy. We will quantitatively analyze this performance drop in the experimental results (Sec. VI-B2a).

While our primary analysis focuses on single-channel failure scenarios, we additionally extend our evaluation to simultaneous dual microphone failure conditions in Sec. VI-C, identifying the practical boundaries of the proposed framework. We specifically target a 4-channel microphone array configuration, which is widely adopted in robotic auditory perception [1], [2], [26]. This choice is motivated by recent deep learning advancements demonstrating that, even with this compact setup, models can achieve exceptional localization precision, reaching a MAE as low as 2.65° [3]. However, maintaining at least three operational microphones is the fundamental geometric prerequisite to satisfy the constraint ($M \geq 3$) defined in Sec. III-A, ensuring unique 2D localization without front-back ambiguity. Scenarios involving two or more failed microphones generally result in insufficient spatial information for reliable full 360° DOA estimation. Consequently, we evaluate the proposed method under the single-fault assumption, which represents the minimal viable configuration required to sustain the robust performance of the target platform.

IV. FAULT-RESILIENT SOUND SOURCE LOCALIZATION FRAMEWORK

In this section, we propose a Fault-Resilient SSL framework that minimizes localization accuracy degradation under abrupt hardware failures. As illustrated in Fig. 3, the system operates as a sequential pipeline comprising two core modules: 1) a **Microphone Fault Diagnosis** module driven by a lightweight time-frequency network, and 2) a **TTA** module based on confidence-aware self-training. Crucially, the proposed framework is not a simple combination of fault diagnosis and TTA, but a unified system in which diagnosed fault semantics directly determine the form of intervention and adaptation.

A. LIGHTWEIGHT AND RELIABLE FAULT DIAGNOSIS

Since the proposed framework is intended for deployment on resource-constrained edge platforms, the fault diagnosis module must be computationally efficient while maintaining high accuracy. Complex architectures such as Transformers or deep CNNs often entail high computational costs, which are unnecessary for the relatively distinct patterns of sensor faults. Therefore, we propose a **Lightweight Frequency-Time Projection network** (Fig. 4) that minimizes learnable

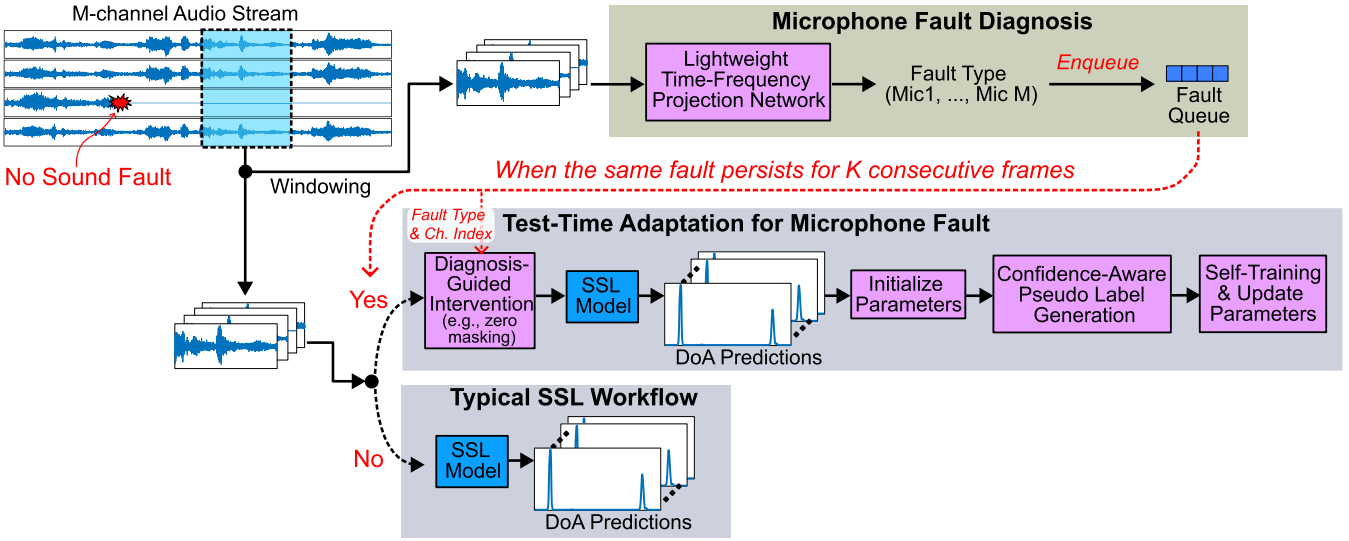


FIGURE 3. Overview of the proposed Fault-Resilient Sound Source Localization (SSL) framework. The system processes the M-channel audio stream using two cascaded modules: Microphone Fault Diagnosis and test-time adaptation (TTA). First, a Lightweight Time-Frequency Projection Network identifies fault types, using a Fault Queue to filter transient prediction errors. The TTA process is triggered only when a consistent fault persists for K consecutive frames. Crucially, the adaptation is diagnosis-guided: the identified fault type and channel index are utilized to apply targeted interventions (e.g., explicit zero-masking for signal replication) before feeding features to the SSL model. Subsequently, the model estimates a sequence of direction-of-arrival (DOA) likelihoods to capture the source trajectory. These predictions are then used to generate confidence-aware hard pseudo-labels for self-training.Unlike the typical fixed-weight workflow (bottom), our framework dynamically updates the model parameters to recover performance in faulty scenarios.

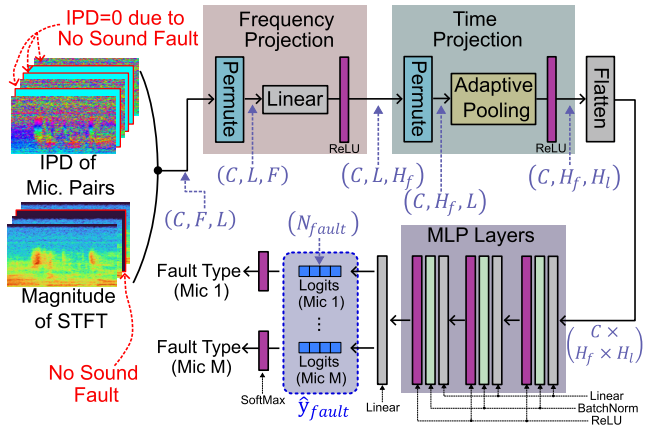


FIGURE 4. Architecture of the proposed Lightweight Frequency-Time Projection Network for microphone fault diagnosis. The network integrates spectral cues (magnitude of short-time Fourier transform, STFT) and spatial cues (inter-channel phase difference, IPD) to robustly identify diverse failure modes. To ensure computational efficiency, the model sequentially projects the frequency dimension ($F \rightarrow H_f$) via a linear layer and compresses the time dimension ($L \rightarrow H_l$) via adaptive pooling. The aggregated features are finally processed by a multi-layer perceptron (MLP) classifier to determine the specific fault type $\hat{y}_{fault}$ for each microphone.

parameters by decoupling frequency and time domain processing.

The network takes a concatenated feature tensor $\mathbf{X}_t \in \mathbb{R}^{C\times L\times F}$ as input at the t-th time frame. For notational simplicity, we omit the time frame index t in the subsequent descriptions. Here, C represents the total number of feature channels, and L and F denote the number of time bins and frequency bins, respectively. Specifically, the channel dimension C is constructed by stacking the M channels of STFT magnitude spectra and $M(M-1)/2$ channels of IPD features computed from all possible microphone pairs. Notably, the inclusion of IPD features is essential for identifying phase-related anomalies such as *Signal Replication*. Unlike energy-based features (e.g., VAD [10] or magnitude) which often fail to distinguish replicated signals from normal speech due to preserved energy levels, IPD explicitly captures the inter-channel spatial correlations, enabling the detection of unnatural phase consistency.

To effectively reduce the model size, we process the frequency and time dimensions sequentially. First, the **Frequency Projection** layer maps the high-dimensional frequency features to a latent space of size H_f. Let $\mathbf{f}_{c,l} \in \mathbb{R}^F$ be the frequency feature vector for channel c at time step l. The projection is defined as:

$$\mathbf{z}_{c,l} = \sigma(\mathbf{W}_f \mathbf{f}_{c,l} + \mathbf{b}_f), \tag{5}$$

where $\mathbf{W}_f \in \mathbb{R}^{H_f\times F}$ and $\mathbf{b}_f$ are learnable weights and biases, and $\sigma(\cdot)$ is the ReLU activation function. Crucially, $\mathbf{W}_f$ is shared across all time steps l and channels c, allowing the model to extract consistent spectral features independent of temporal variations while significantly reducing parameter count.

Subsequently, to efficiently handle the temporal dimension L, we employ a **Time Projection** layer utilizing adaptive pooling. Instead of processing the entire temporal sequence with high-dimensional fully connected layers, this layer compresses the temporal dimension L into a compact fixed size H_l. This operation is mathematically formulated as computing the local average for the ℓ-th output index ($0 \le \ell < H_l$):

$$\mathbf{h}_{c,\ell} = \frac{1}{e_\ell - s_\ell} \sum_{j=s_\ell}^{e_\ell - 1} \mathbf{z}_{c,j}, \tag{6}$$

where the start index s_ℓ and end index e_ℓ are determined based on the input length L:

$$s_\ell = \left\lfloor \frac{\ell \cdot L}{H_l} \right\rfloor, \quad e_\ell = \left\lfloor \frac{(\ell+1) \cdot L}{H_l} \right\rfloor. \tag{7}$$

This formulation serves as a parameter-free dimensionality reduction technique that aggregates temporal information into a consistent representation $\mathbf{H} \in \mathbb{R}^{C \times H_l \times H_f}$. The validity of this temporal compression lies in the stationarity of sensor faults; typically, hardware failures (as illustrated in Fig. 2) exhibit consistent patterns within a short observation window. Therefore, aggregating global temporal statistics via pooling effectively captures the fault signature while minimizing learnable parameters.

Finally, the aggregated features are flattened and passed through a multi-layer perceptron (MLP) classifier to produce the fault logits $\hat{y}_{fault} \in \mathbb{R}^{M \times N_{fault}}$, where N_{fault} denotes the number of predefined fault types (including the normal state). Thanks to this decoupled design, our model requires only **3.29 million parameters**. In contrast, a naive MLP approach processing the raw flattened input would require approximately **265 million parameters**, representing a reduction of over **98.7%** (see Sec. VI-A2 for detailed analysis). Despite its simplicity, our experimental results demonstrate that this lightweight model achieves a detection accuracy of **99.97%** in random single-fault scenarios, proving that heavy architectures are not required for this task.

The network is trained in a supervised manner by minimizing the cross-entropy loss between the predicted probabilities and the ground-truth fault labels. To ensure robust generalization across diverse acoustic conditions, we utilized an existing clean speech dataset and synthetically injected the microphone fault patterns defined in Section III-B (e.g., *No Sound, Attenuation, Impulsive Noise, Continuous Noise, and Signal Replication*). This data synthesis strategy allows the model to learn distinct spectral and spatial signatures of hardware failures without requiring expensive real-world faulty data collection. Comprehensive details regarding the data generation pipeline and training protocols are provided in Sec. V-A.

Although the proposed model achieves near-perfect accuracy (99.97% as described in Sec. VI-A2), minimizing sporadic false predictions remains crucial for maintaining system efficiency and reliability. For example, a *false positive* (diagnosing a healthy microphone as faulty) would trigger a redundant adaptation process, incurring unnecessary computational overhead. Therefore, prioritizing the robustness of the decision is paramount. To address this, we integrate the **Fault Queue** mechanism introduced in Sec. IV as a post-processing step (Fig. 3). The predicted fault label for the current frame is pushed into a first-in-first-out queue of length K. The system confirms a hardware failure and triggers the TTA process only when the queue is fully populated with an identical fault type (i.e., the same fault persists for K consecutive frames). This temporal consistency check ensures that the system reacts only to sustained hardware malfunctions, effectively filtering out transient outliers.

B. DIAGNOSIS-GUIDED CONFIDENCE-AWARE TTA

Once a sustained microphone fault is confirmed by the Microphone Fault Diagnosis module (Sec. IV-A), the system triggers the TTA process. The overall workflow of the proposed TTA framework is illustrated in Fig. 5. Leveraging the probabilistic output representation defined in Sec. III-A, our framework adapts the pre-trained model using its own predictions. As discussed in Sec. III-A, this architectural choice—predicting a spatial likelihood distribution rather than a single coordinate—is the fundamental prerequisite that allows us to quantify prediction uncertainty and enable unsupervised adaptation.

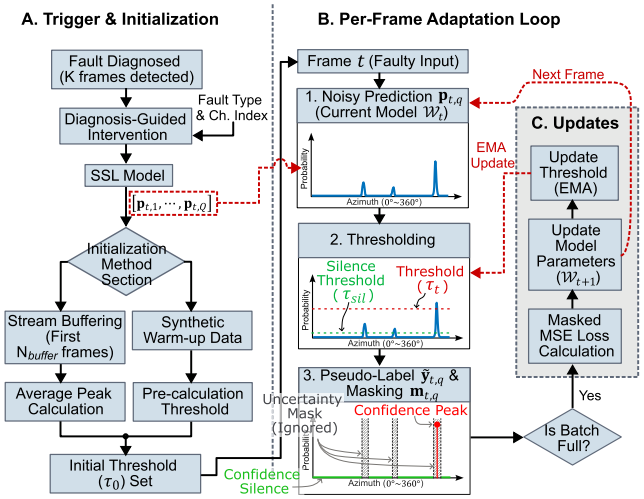


FIGURE 5. Flowchart of the proposed test-time adaptation (TTA) process.(A) Upon fault triggering, diagnosis-guided intervention (e.g., explicit zero-masking) is selectively applied to the input data depending on the identified fault type. Subsequently, the initial confidence threshold τ_0 is set via either stream buffering (N_{buffer} frames) or immediate synthetic warm-up.(B) In the adaptation loop, the system generates sequences of hard pseudo-labels $\tilde{\mathbf{y}}_{t,q}$ and uncertainty masks $\mathbf{m}_{t,q}$ by filtering the direction-of-arrival (DOA) likelihoods $\mathbf{p}_{t,q}$. This step uses dynamic confidence (τ_t) and silence (τ_{sil}) thresholds to retain reliable predictions while discarding ambiguous or silent predictions.(C) Finally, the model parameters $\mathcal{W}_t$ and threshold τ_t are updated on a mini-batch basis via sequence-wise masked mean squared error (MSE) loss and exponential moving average (EMA), respectively.

For the adaptation strategy, we adopt a pseudo-labeling approach over entropy minimization techniques (e.g., TENT [14]). While entropy minimization effectively sharpens prediction distributions, pseudo-labeling provides explicit supervision signals, generally yielding superior performance in recovering from severe domain shifts, as discussed in Sec. VI-B2c. However, the efficacy of this approach hinges critically on the quality of the generated labels. Predictions derived from a compromised microphone array are inherently noisy and unstable (Fig. 6).

To mitigate this risk and ensure stability, we propose a **Confidence-Aware Pseudo-Labeling** strategy. By strictly filtering out low-confidence predictions using a dynamic

threshold, this mechanism ensures that the model is updated only with high-quality supervision signals, effectively preventing the error accumulation that typically arises from training on noisy predictions. Furthermore, while the proposed confidence-aware mechanism is effective, there is room to further enhance adaptation performance in severe fault scenarios that generate deceptive spatial cues (e.g., Signal Replication). To maximize robustness in such challenging cases, we incorporate a **Diagnosis-Guided Intervention** strategy (Sec. IV-B4), which leverages specific diagnosis results to enable adaptation performance beyond the capabilities of unsupervised filtering alone.

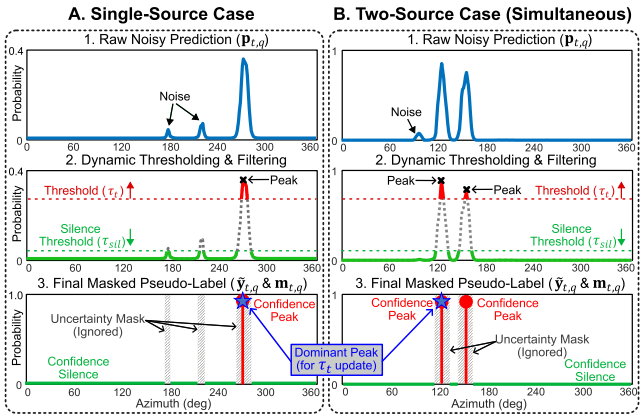


FIGURE 6. **Visualization of Confidence-Aware Hard Pseudo-Label generation for (A) Single-Source and (B) Two-Source scenarios at a specific intra-frame step q. First, the raw direction-of-arrival (DOA) likelihoods $\mathbf{p}_{t,q}$ (top) are evaluated against the dynamic peak threshold τ_t and silence threshold τ_{sil} (middle). Crucially, ambiguous predictions that fail to meet the confidence criteria are explicitly masked out (Uncertainty Mask) to prevent error propagation. Consequently, the final pseudo-labels $\tilde{\mathbf{y}}_{t,q}$ (bottom) assign 1 only to distinct Confidence Peaks and 0 to Confidence Silence, ensuring that only reliable spatial cues are used for adaptation by masking out ambiguous regions using $\mathbf{m}_{t,q}$. Among all valid Confidence Peaks, the highest-confidence peak is designated as the Dominant Peak, which is exclusively used for updating the dynamic threshold τ_t via the EMA update rule (Eq. 11). In the Two-Source case (B), where multiple peaks exceed τ_t, all peaks contribute to the pseudo-label targets, while only the single Dominant Peak governs the threshold update, ensuring stable and consistent threshold tracking regardless of the number of active sources.**

1) CONFIDENCE-AWARE PSEUDO-LABEL GENERATION

The core intuition behind our approach is that the model's prediction confidence serves as a reliable proxy for localization accuracy, even in the presence of microphone faults. We observe that even with corrupted audio inputs, the SSL model often produces high-likelihood predictions (i.e., sharp peaks) when the underlying spatial cues remain distinguishable, as visualized in Fig. 6. In such cases, the model is considered to be confident, and the corresponding DOA estimation is likely to be accurate. Building on this insight, we distinguish between high-confidence regions and ambiguous predictions to construct a robust training target.

To implement this, we generate a **Confidence-Aware Hard Pseudo-Label sequence** $\tilde{\mathbf{y}}_{t,q}$ and a binary mask sequence $\mathbf{m}_{t,q} \in \{0,1\}^K$ for each intra-frame step $q = 1, \ldots, Q$ by strictly filtering the sequential model predictions (Fig. 5-B). We define three distinct confidence regions based on a time-varying **Dynamic Confidence Threshold** τ_t and a fixed **Silence Threshold** τ_{sil}. Based on this classification, we construct the pseudo-labels and masks for each step q as follows:

1) **Confidence Peak (Positive Target):** First, we identify the set of local maxima indices $\mathcal{K}^*_{t,q}$ from the predicted likelihood vector $\mathbf{p}_{t,q}$ at step q. To filter out spurious peaks caused by faults, we strictly accept a peak as a reliable source estimate only if its probability exceeds the dynamic threshold τ_t. For these indices, we assign a positive pseudo-label ($\tilde{\mathbf{y}}_{t,q,k} = 1$) and activate the mask ($\mathbf{m}_{t,q,k} = 1$).
2) **Confidence Silence (Negative Target):** Angular bins with probabilities falling below the fixed silence threshold τ_{sil} are treated as confident non-source regions. For these indices, we assign a negative pseudo-label ($\tilde{\mathbf{y}}_{t,q,k} = 0$) and activate the mask ($\mathbf{m}_{t,q,k} = 1$) to suppress background noise.
3) **Uncertainty Region (Masked):** Any angular bins that do not fall into either the *Confidence Peak* or *Confidence Silence* categories are defined as the Uncertainty Region. This implies that bins exhibiting intermediate probability ($\tau_{sil} \le \mathbf{p}_{t,q,k} \le \tau_t$) or those with high probability that are not local maxima (e.g., slopes of a peak) are treated as ambiguous. Crucially, these uncertain regions are masked out ($\mathbf{m}_{t,q,k} = 0$), effectively excluding them from the loss calculation to prevent error accumulation.

Based on these definitions, we construct the hard pseudo-label $\tilde{\mathbf{y}}_{t,q}$ and the mask $\mathbf{m}_{t,q}$ using a unified logic (visualized in the bottom row of Fig. 6-A and B):

$$(\tilde{\mathbf{y}}_{t,q,k}, \mathbf{m}_{t,q,k}) = \begin{cases} (1,1), & \text{if } k \in \mathcal{K}^*_{t,q} \text{ and } \mathbf{p}_{t,q,k} > \tau_t \\ \quad (\textit{Confidence Peak}), & \\ (0,1), & \text{if } \mathbf{p}_{t,q,k} < \tau_{sil} \\ \quad (\textit{Confidence Silence}), & \\ (0,0), & \text{otherwise} \\ \quad (\textit{Uncertainty Mask}), & \end{cases} \tag{8}$$

where $\mathcal{K}^*_{t,q}$ denotes the set of local maxima indices for the q-th intra-frame step.

This tripartite labeling scheme effectively mitigates the significant degradation in SSL estimation performance caused by microphone faults. By employing Confidence Peak, we enforce prediction sharpening to recover clear decision boundaries, while the Uncertainty Mask and Confidence Silence act as safety mechanisms to prevent confirmation bias from ambiguous or noisy inputs. Collectively, these components enable the model to robustly adapt to hardware

failures, leading to a significant recovery in SSL accuracy. We quantitatively verify the contribution of each component and the overall performance improvement in the experimental results (Sec. VI-B).

2) SELF-TRAINING OBJECTIVE

Using the generated hard pseudo-labels and uncertainty masks, we formulate the optimization objective for TTA. Consistent with standard TTA protocols [14], we adopt a mini-batch-based optimization strategy rather than updating the model at every single time step. Specifically, the system accumulates the input data and their corresponding pseudo-labels over a sequence of B consecutive frames. Once a batch is fully populated, the model parameters $\mathcal{W}_t$ are updated by minimizing the sequence-wise masked MSE loss averaged over the batch:

$$\mathcal{L}_{batch} = \frac{1}{B \cdot Q} \sum_{i=1}^{B} \sum_{q=1}^{Q} \left(\frac{1}{V_{i,q}} \sum_{k=1}^{K} \mathbf{m}_{i,q,k} \left(\tilde{\mathbf{y}}_{i,q,k} - \mathbf{p}_{i,q,k} \right)^2 \right), \quad (9)$$

where $V_{i,q} = \sum_{k=1}^{K} \mathbf{m}_{i,q,k} + \epsilon$ represents the number of valid angular bins for the q-th step of the i-th frame, serving as a normalization factor and ϵ is a small constant to prevent division by zero. As mentioned in Sec. III-A, we adopt MSE loss following prior SSL formulations that treat spatial likelihood estimation as a regression over angular bins.

While reliance on mini-batch statistics—particularly for batch normalization—can theoretically introduce sensitivity to batch size, our empirical analysis in Sec. VI-B6 demonstrates that the proposed framework maintains high stability across varying batch sizes. Based on this verification that a compact batch size is sufficient for robust adaptation, we set $B = 8$ for our experiments. To minimize the objective, we employ the Adam optimizer with a learning rate of $\eta = 1 \times 10^{-5}$, which was empirically found to strike an optimal balance between adaptation speed and stability.

3) ADAPTIVE THRESHOLD ESTIMATION

To ensure the robustness of the labeling process, maintaining an appropriate confidence threshold τ_t is paramount. As empirically demonstrated in Sec. VI-B4, employing a static threshold is insufficient for two key reasons. First, the optimal decision boundary varies significantly across different fault types (e.g., lower for signal replication, higher for white noise). Second, the model's confidence is not constant but evolves continuously as the parameters are updated via self-training. A fixed threshold lacks the flexibility to adapt to this evolution, potentially causing false negatives in the early stages or admitting noise as the model sharpens. To address these challenges and synchronize the selection criteria with the model's evolving state, we employ an adaptive mechanism driven by an exponential moving average (EMA).

a: ADAPTIVE UPDATE MECHANISM

To robustly track the confidence trends while accommodating scenarios where the number of active sound sources is unknown, we calculate a representative confidence score $\bar{p}_{batch}$ based on the **Dominant Peak Assumption**. Specifically, if the maximum probability in a prediction step exceeds the silence threshold τ_{sil}, we assume the presence of at least one valid sound source. Accordingly, we extract the maximum probability value from each valid step (i, q) within the current mini-batch. The representative confidence $\bar{p}_{batch}$ is then computed as the average of these dominant peaks across all valid steps in the batch:

$$\bar{p}_{batch} = \frac{1}{|\mathcal{I}_{valid}|} \sum_{(i,q) \in \mathcal{I}_{valid}} \max(\mathbf{p}_{i,q}), \quad (10)$$

where $\mathcal{I}_{valid} = \{(i, q) \mid i \in [1, B], q \in [1, Q], \max(\mathbf{p}_{i,q}) > \tau_{sil}\}$ denotes the set of index pairs for steps containing valid spatial cues.

Finally, the dynamic threshold is updated using the batch-aggregated confidence to ensure stability. Following the design principles in [43], we set the smoothing factor λ to a low value (e.g., $\lambda = 0.1$) to suppress rapid fluctuations. Crucially, if no reliable peaks are detected in the entire batch (i.e., $\mathcal{I}_{valid} = \emptyset$), the update is skipped to prevent the threshold from drifting towards noise levels:

$$\tau_{t+1} = \begin{cases} (1 - \lambda)\tau_t + \lambda \bar{p}_{batch}, & \text{if } \mathcal{I}_{valid} \neq \emptyset \\ \tau_t, & \text{otherwise.} \end{cases} \quad (11)$$

While this Dominant Peak Assumption does not explicitly model multiple concurrent sources, we empirically observe that the highest-confidence peak remains a reliable indicator of overall prediction confidence under microphone fault conditions.

b: THRESHOLD INITIALIZATION

Determining an appropriate initial threshold τ_0 plays a pivotal role in boosting the adaptation performance (Fig. 5-A). To ensure a stable starting point for the EMA update, we calculate τ_0 by averaging the representative confidence scores (dominant peaks) observed within an initialization buffer of N_{buffer} frames. Crucially, to prevent low-confidence noise from biasing the initialization, we aggregate confidence scores from all valid intra-frame steps where the maximum probability exceeds the silence threshold:

$$\tau_0 = \frac{1}{|\mathcal{I}_{init}|} \sum_{(t,q) \in \mathcal{I}_{init}} \max(\mathbf{p}_{t,q}), \quad (12)$$

where $\mathcal{I}_{init} = \{(t, q) \mid t \in [1, N_{buffer}], q \in [1, Q], \max(\mathbf{p}_{t,q}) > \tau_{sil}\}$ denotes the set of index pairs for steps containing valid spatial cues within the initialization buffer.

This data-driven initialization sets the EMA to a value tailored to the specific fault type (e.g., lower for Signal Replication, higher for White Noise). As empirically demonstrated in Sec. VI-B4, this strategy prevents the cold start problem

and boosts TTA performance by ensuring that the model updates are stable from the very first step.

Regarding the buffer size, calculating τ_0 from a live data stream introduces a latency proportional to N_{buffer}. However, our sensitivity analysis in Sec. VI-B3 indicates that a short window of $N_{buffer} = 8$ frames is sufficient to estimate a reliable τ_0 comparable to using larger buffers (e.g., 64 frames). Based on this finding, we set $N_{buffer} = 8$, resulting in an initial latency of approximately 8 seconds under our experimental configuration. Alternatively, for scenarios requiring zero latency, a *Synthetic Warm-up* strategy can be employed using pre-calculated statistics from the synthetic faulty dataset (Dataset A), which was originally generated for training the Fault Diagnosis module (Sec. V-A2).

c: STATIC SILENCE THRESHOLD

In contrast to the dynamic peak threshold, we maintain a fixed value for the silence threshold τ_{sil}. This design choice is grounded in the intrinsic behavior of the pre-trained SSL models, which are optimized to assign near-zero likelihoods to non-source regions during training. Furthermore, unlike active source signatures which are typically distorted or shifted by hardware faults, the acoustic characteristic of silence (i.e., near-zero input energy) remains largely invariant. Given this stability, the definition of silence does not require adaptation. Instead, it serves as an absolute anchor to prevent the model from drifting into instability by suppressing background noise. Therefore, we fix $\tau_{sil} = 0.02$, a value empirically determined to be sufficiently low to guarantee non-source regions while preserving potential weak sources (detailed analysis in Sec. VI-B5).

4) DIAGNOSIS-GUIDED INTERVENTION STRATEGY

While the proposed TTA framework effectively recovers performance across most fault scenarios, certain severe faults—most notably *Signal Replication*—present a persistent challenge. Even with the robust dynamic thresholding described above, the recovery in these specific cases can remain suboptimal compared to other fault types. This is primarily because *Signal Replication* creates deceptive spatial cues that mimic valid speech characteristics, making it intrinsically difficult for the model to distinguish between the true source and the hardware-induced ghost source purely through self-supervision.

To overcome this limitation and further boost adaptation performance, we leverage the precise fault classification provided by our system to implement a **Diagnosis-Guided Intervention** strategy (Fig. 5-A). Unlike general TTA frameworks that adapt blindly to domain shifts, our approach exploits the explicit knowledge of the fault type and the affected channel index to apply targeted pre-processing before the data enters the adaptation loop.

Specifically, for the *Signal Replication* fault, we intervene by explicitly forcing the input of the diagnosed faulty channel to zero (i.e., channel masking). This structural intervention breaks the erroneous phase relationships caused by the replicated signal, effectively converting a complex, deceptive fault into a simpler *dead channel (No Sound)* scenario that the SSL model can handle more robustly. The efficacy of this diagnosis-guided intervention is quantitatively validated in Sec. VI-B2e, where we demonstrate that removing the deceptive artifacts leads to a substantial performance boost in scenarios where the proposed TTA framework struggles. More broadly, this result highlights that incorporating explicit fault semantics can fundamentally reshape the adaptation landscape, enabling recovery beyond the limits of data-driven adaptation alone.

V. EXPERIMENTS

A. FAULT DATA SYNTHESIS ON REAL-WORLD AUDIO

To validate the robustness of the proposed **Fault-Resilient Sound Source Localization Framework**—which integrates microphone fault diagnosis with TTA—in realistic acoustic environments, we utilize the SSLR dataset [1] as our base corpus. This dataset consists of real-world multichannel audio recordings captured using a robot equipped with a 4-channel circular microphone array. Specifically, the recordings feature human speech generated by live talkers as well as loudspeakers playing back recorded speech. Crucially, the dataset encompasses varying acoustic complexities, ranging from single-source scenarios to cases with two overlapped sound sources, captured in diverse indoor environments with varying levels of background noise and reverberation.

To simulate realistic failure scenarios, we synthetically inject the five fault types defined in Sec. III-B into the clean SSLR recordings.We specifically evaluate the system under single-channel failure conditions, simulating the critical state where the array operates with the minimum number of sensors ($M = 3$). For a detailed discussion on the geometric constraints and the rationale behind this single-fault assumption, please refer to Sec. III-B.

While the primary evaluation focuses on single-channel failure conditions, real-world deployments may encounter simultaneous failures of multiple microphones. In particular, when two microphones simultaneously lose all spatial information (e.g., No Sound or Signal Replication), the geometric constraints for reliable 2D DOA estimation can no longer be satisfied, leading to severe localization degradation. To investigate how the proposed Microphone Fault Diagnosis model and TTA framework behave under such challenging conditions, we extend our evaluation to a simultaneous dual microphone failure scenario in Sec. VI-C.

1) FAULT INJECTION SCENARIOS

To systematically evaluate the impact of hardware failures, we synthetically inject microphone faults into the clean SSLR recordings. Adopting the mathematical fault model defined in Eq. 4 (Sec. III-B), we generate corrupted signals by manipulating the gain factor $\alpha_m(n)$ and the additive noise term $\beta_m(n)$. Representative waveform examples for these synthesized fault signals are visualized in Fig. 2. The

specific implementation parameters for each fault scenario are configured as follows:

- **No Sound (Dead Channel):** We strictly set both the gain and noise terms to zero ($\alpha_m(n) = 0, \beta_m(n) = 0$), resulting in a completely silent channel containing no acoustic information.
- **Attenuation (Gain Mismatch):** Simulating severe sensitivity degradation based on the criteria in [10], we randomly sample the gain factor $\alpha_m(n)$ from a uniform distribution $U(0.02, 0.05)$, corresponding to a 95% to 98% reduction in signal amplitude, with no additive noise ($\beta_m(n) = 0$).
- **Impulsive Noise (Pops & Clicks):** To model intermittent spikes, we set $\alpha_m(n) = 1$ and inject sparse impulsive noise $\beta_m(n)$. The impulses are generated with random amplitudes in the range of $\pm[0.3, 1.0]$ at a frequency of 10 to 20 occurrences per second.
- **Continuous Noise (White Noise):** We inject Gaussian white noise $\beta_m(n) \sim \mathcal{N}(0, \sigma^2_{fault})$ while maintaining the original signal ($\alpha_m(n) = 1$). To simulate varying intensities of interference, the noise variance σ^2_{fault} is calibrated to achieve a Signal-to-Noise Ratio (SNR) randomly sampled from the range of $[-5, 10]$ dB relative to the power of the original speech signal.
- **Signal Replication (Cross-Channel Copy):** As defined in Sec. III-B, we replace the content of the target microphone m with the signal from another randomly selected microphone j ($j \neq m$), such that $\tilde{x}_m(n) = x_j(n)$. This creates a valid speech signal that lacks the correct spatial phase information.

2) CONSTRUCTION OF EXPERIMENTAL DATASETS

Using the fault injection protocols described above, we generate two distinct datasets derived from the SSLR corpus, each serving a specific purpose in our evaluation pipeline. To ensure experimental integrity, we strictly adhere to the original train/test splits of the SSLR dataset, preventing any data leakage between the diagnosis model training (Dataset A) and the TTA evaluation (Dataset B).

Dataset A (For Fault Diagnosis): This dataset is constructed to train and evaluate the **Microphone Fault Diagnosis** module (Sec. IV-A). To ensure the diagnosis model generalizes across different sensors and failure modes, we apply a randomized injection strategy. For each audio sample in the SSLR corpus, we randomly select one target microphone channel and apply one of six conditions: the five fault types defined in the previous section or a *No Fault (Normal)* condition. The inclusion of the *No Fault* class is crucial, as it enables the diagnosis model to discriminate between actual hardware failures and normal acoustic variations.

Dataset B (For TTA): This dataset is designed to simulate a realistic deployment scenario where the system must sequentially diagnose a hardware failure and recover performance via TTA (Sec. IV-B). Unlike the randomized nature of Dataset A, Dataset B assumes a persistent failure on a specific sensor during robot operation. Specifically, we fix the fault location to the last microphone ($m = 4$) of the SSLR test split and generate five independent evaluation sets, each corresponding to one of the five fault types. This controlled setup allows us to isolate the impact of each specific fault type and quantitatively verify whether our framework—after correctly diagnosing the faulty channel using the model trained on Dataset A—can successfully recover localization accuracy through adaptation.

3) REAL-ENVIRONMENT DATASET UNDER MICROPHONE FAULT

To verify that the proposed framework validated on the synthetically injected fault scenarios generalizes to authentic hardware failures, we additionally collect a real-environment dataset by physically reproducing a *No Sound* fault condition using a 4-channel microphone array with a different geometry from that used in the SSLR dataset. To accommodate this array configuration mismatch, we generate a synthetic dataset using a sound simulator (Pyroomacoustics [44]) and dry speech signals from the NIGENS dataset [45], and use it to fine-tune both the DOA estimation model and the proposed Microphone Fault Diagnosis model for the target array. A detailed description of the data collection procedure and fine-tuning protocol is provided in Sec. VI-D.

B. IMPLEMENTATION DETAILS

To ensure the reproducibility of our experiments, we detail the specific configurations for audio preprocessing, model architectures, and training hyperparameters. All models and experiments were implemented using the PyTorch framework. Training and evaluation were conducted on a workstation equipped with an Intel(R) Core(TM) Ultra 9 285K CPU and a single NVIDIA RTX 5090 GPU.

Audio Preprocessing: All audio recordings in the SSLR dataset were resampled to 24 kHz. The input for both the localization and diagnosis modules consists of 1-second non-overlapping audio segments (24,000 samples). For the spectral feature extraction used in the Fault Diagnosis module, we compute the STFT to generate time-frequency features with dimensions $F = 257$ (frequency bins) and $L = 50$ (time frames).

Backbone SSL Models: To evaluate the generalizability of our framework, we employ three distinct SSL architectures: **MLP-GCC** [1], **GCC-PHAT-SM** [2], and **MPT-HiFTA** [3]. We adopt the exact network configurations proposed in their respective original papers. In this work, these models process a 1-second input frame to predict a sequence of likelihood vectors corresponding to $Q = 10$ intra-frame steps. Consequently, the output dimension is defined as $Q \times K$ (where $K = 360$). Prior to TTA, these models are pre-trained on the clean (fault-free) SSLR training set to serve as the initial baseline ($f_{\mathcal{W}_0}$).

Microphone Fault Diagnosis Model: For the proposed Lightweight Frequency-Time Projection network (Fig. 4), we set the projection dimensions to $H_f = 32$ and $H_l = 8$. The final classifier is composed of three MLP hidden layers with sizes [1024, 512, 256]. The model is trained on Dataset A for 50 epochs using the Adam optimizer with a learning rate of 1×10^{-3}. Thanks to the lightweight architecture, the inference latency is extremely low, averaging 0.099 ms on a CPU and 0.129 ms on a GPU per frame. This negligible computational cost fully guarantees real-time performance with minimal overhead. During inference, the Fault Queue length is set to $N = 2$, meaning a fault must persist for at least two consecutive frames (2 seconds) to trigger the adaptation process.

TTA Protocols: Following the TTA strategy formulated in Sec. IV-B2, we set the mini-batch size to $B = 8$ for all experiments. For evaluation, we strictly adhere to the standard continuous TTA protocol: the localization performance is measured on the incoming mini-batch before any parameter updates are applied. Subsequently, the model is updated using the self-training objective based on the current batch, ensuring that the reported metrics reflect the system's real-time adaptation capability under streaming data.

Regarding the adaptive initialization of the dynamic threshold, we estimate τ_0 using the first $N_{buffer} = 8$ samples of the incoming test stream. Other TTA-specific hyperparameters, such as the learning rate, EMA momentum, and silence threshold, are configured exactly as specified in Sec. IV-B.

Computational Overhead of TTA. We further quantify the computational cost induced by the adaptation process, defined as the additional time required for pseudo-label generation and parameter updates (excluding the forward inference pass). When executed on a GPU, the average overhead per frame is marginal: 3.11 ms for MPT-HiFTA, 1.04 ms for GCC-PHAT-SM, and 0.73 ms for MLP-GCC. Even in a CPU-only environment, the overheads remain highly manageable at 48.77 ms, 2.91 ms, and 2.12 ms, respectively. As will be discussed in Sec. VI-B, although larger models like MPT-HiFTA incur higher latency, they yield superior DOA estimation accuracy and more stable adaptation trajectories. Crucially, even the maximum overhead observed ($\approx$ 48.77 ms on CPU) is negligible compared to the input frame duration (1 second), confirming that our TTA framework operates well within real-time constraints regardless of the hardware platform.

VI. RESULT

A. MICROPHONE FAULT DIAGNOSIS

We first evaluate the performance of the proposed Microphone Fault Diagnosis module. To ensure a rigorous evaluation, all models were trained using the training split of **Dataset A**, and the quantitative results reported in Table 1 were obtained from the held-out test split. The table summarizes the classification accuracy across different methods, comparing our proposed Lightweight Frequency-Time Network (**Lighweight MLP**) against three baselines: the traditional VAD-based thresholding method (**VAD-Threshold**) [10], a VAD-based MLP (**VAD-MLP**), and a **Heavyweight MLP** processing raw STFT and IPD features.

1) LIMITATIONS OF VAD-BASED APPROACHES

As observed in Table 1, the traditional **VAD-Threshold** method exhibits significant limitations, achieving an overall accuracy of only 52.70% (standard deviation: 46.83%). This performance drop is expected, as the original algorithm [10] was not designed to account for complex fault types such as *Continuous Noise* (0.00%) and *Signal Replication* (0.00%). To investigate whether VAD features themselves contain sufficient information to distinguish these uncovered faults—beyond the limitations of the original thresholding logic—we implemented a deep learning-based variant (**VAD-MLP**) by replacing the manual thresholds with an MLP classifier. While this data-driven extension significantly improves the detection of *Continuous Noise* (99.61%) by learning non-linear patterns, it still struggles with *Signal Replication*, yielding a low accuracy of 70.84%. This finding confirms that magnitude-derived VAD features alone are fundamentally insufficient for identifying phase-related hardware failures, even when processed by a powerful neural classifier.

2) ROBUSTNESS OF THE PROPOSED METHOD

In contrast, models utilizing spatial features (STFT + IPD) demonstrate superior robustness. Both the **Heavyweight MLP** and our **proposed Lightweight MLP** achieve near-perfect overall accuracy ($>$ 99.1%) with remarkably low standard deviations (1.87% and 0.06%, respectively), indicating not only high mean performance but also highly consistent diagnosis accuracy across all fault types. Notably, the proposed **Lightweight MLP** outperforms the Heavyweight counterpart in distinguishing transient anomalies. Specifically, for *Impulsive Noise (Pops & Clicks)*, the proposed method achieves 99.83% accuracy compared to 95.35% for the Heavyweight model. This improvement suggests that our Frequency-Time Projection architecture efficiently captures the spectro-temporal signatures of intermittent spikes without the need for massive parameterization. Moreover, the proposed method successfully identifies the challenging *Signal Replication* fault with 99.95% accuracy, proving that the inclusion of IPD features effectively resolves phase inconsistencies that baffle traditional methods. These results validate that our lightweight diagnosis module provides a highly reliable trigger for the subsequent TTA process.

B. TTA FOR MICROPHONE FAULT

1) RELIABILITY VERIFICATION OF THE DIAGNOSIS MODULE

Before analyzing the localization performance recovery via TTA, we first verify the reliability of the proposed Microphone Fault Diagnosis module on **Dataset B**. Unlike

TABLE 1. Quantitative comparison of fault diagnosis performance across different methods on Dataset A. While the VAD-based baseline fails to detect complex faults such as *Continuous Noise* and *Signal Replication*, the proposed lightweight model effectively leverages spectral and spatial features (STFT+IPD) to achieve near-perfect accuracy across all fault types. The Overall Accuracy is reported as mean ± standard deviation across all fault types. The best performance for each metric is highlighted in bold.

Method	Input Features	Overall Acc. (%)	Class-wise Diagnosis Accuracy (%)					
			No Fault	No Sound	Signal Attenuation	Impulsive Noise	Continuous Noise	Signal Replication
VAD-Threshold [10]	VAD Feats.	52.70 ± 46.83	79.07	**100.0**	**100.0**	37.15	0.00	0.00
VAD-MLP	VAD Feats.	91.76 ± 12.43	99.73	99.93	98.51	81.92	99.61	70.84
Heavyweight MLP	STFT + IPD	99.17 ± 1.87	**99.99**	**100.0**	99.95	95.35	**99.98**	99.73
Proposed (Lightweight MLP)	STFT + IPD	**99.94 ± 0.06**	99.98	**100.0**	99.91	**99.83**	**99.98**	**99.95**

TABLE 2. Impact of the Fault Queue mechanism on diagnosis robustness (Dataset B). While the raw frame-level predictions ($K = 1$) already exhibit high accuracy, employing a minimal queue length of $K = 2$ effectively filters out transient prediction errors. The Overall Accuracy is reported as mean ± standard deviation across all fault types. This results in 100% triggering stability across all fault types, ensuring reliable TTA activation.

Fault Type	Single-Fault Diagnosis Accuracy (%)	
	Frame-level ($K = 1$)	Queue-based ($K = 2$)
No Sound	**100.00**	**100.00**
Attenuation	99.98	**100.00**
Impulsive Noise	99.93	**100.00**
Continuous Noise	**100.00**	**100.00**
Signal Replication	99.92	**100.00**
Average	99.97 ± 0.04	**100.00 ± 0.00**

the randomized transient faults in Dataset A, Dataset B simulates a realistic deployment scenario where a specific hardware failure persists continuously during robot operation. Therefore, reliable triggering—ensuring that the system correctly identifies the fault type and initiates the appropriate adaptation protocol—is a prerequisite for successful TTA.

Table 2 presents the diagnosis accuracy on Dataset B. Even at the single-frame level ($K = 1$), the proposed lightweight model exhibits exceptional performance with an average accuracy of **99.97%** (standard deviation: 0.04%). The minor misclassifications (e.g., 0.08% in *Signal Replication*) correspond to extremely short transition periods or ambiguous frames. However, by applying the proposed **Fault Queue** mechanism with a minimal length of $K = 2$ (i.e., requiring consistency for just two consecutive frames), we effectively filter out these sporadic errors. As shown in the rightmost column of Table 2, this simple temporal smoothing enables 100% classification accuracy across all fault types. This result confirms that our diagnosis module serves as a highly robust trigger for the TTA framework, allowing us to evaluate the adaptation performance with a dependable fault detection baseline.

2) QUANTITATIVE EVALUATION OF TTA PERFORMANCE

Following the reliable fault diagnosis, we evaluate the DOA estimation performance recovery via the proposed TTA framework. To quantify the performance, we employ two standard metrics adopted from [1]: **Mean Absolute Error (MAE)** and **Accuracy**. MAE measures the average absolute deviation between the estimated and ground-truth DOAs, while Accuracy calculates the percentage of estimates where the error falls within a specific tolerance threshold (e.g., 5°). Table 3 summarizes the quantitative results across three SSL models, comparing the baseline (**Before TTA**), the entropy minimization-based TTA (**TENT** [14]), Naive Soft-Labeling (**Soft Pseudo-Label**), and the proposed Hard-Labeling strategy (**Hard Pseudo-Label**) against the **Oracle** benchmark.

a: IMPACT OF FAULTS ON PRE-TRAINED MODELS

As shown in the **Before TTA** column, all fault types induce significant performance degradation compared to the *No Fault* condition, causing the MAE to surge by an average of 238% across all models (from 6.50° to 21.99°). Specifically, *No Sound* and *Signal Replication* cause the most severe errors; for instance, *No Sound* increases the MAE of MPT-HiFTA by 9.20° (from 2.65° to 11.85°), while *Signal Replication* leads to drastic degradation, causing the MAE to surge by 26.18° to reach 28.83°. The former results in a complete loss of information for the affected channel, while the latter distorts the spatial phase relationships inherent in the microphone array geometry, leading to severe confusion in DOA estimation.

For *Attenuation*, the impact varies by model architecture; GCC-based models (MLP-GCC, GCC-PHAT-SM) show relative robustness due to the magnitude-invariant nature of GCC features, whereas MPT-HiFTA, which utilizes raw STFT inputs, suffers a notable decline, with MAE worsening by 9.00° (from 2.65° to 11.65°). Noise-related faults (*Impulsive* and *Continuous Noise*) preserve some original signal characteristics, resulting in relatively smaller but still non-negligible errors (e.g., MAE increases from 2.68° to 5.23° for MLP-GCC and from 2.80° to 5.40° for GCC-PHAT, whereas MPT-HiFTA shows a milder increase ranging from 0.65° to 0.88°).

b: EFFICACY OF HARD PSEUDO-LABELING

The proposed **hard pseudo-labeling** TTA strategy demonstrates universal effectiveness across diverse architectures, successfully mitigating severe domain shifts ranging from structural failures to additive noise. In complex fault

TABLE 3. Quantitative comparison of DOA estimation performance (MAE and Accuracy) across three SSL models under different fault conditions. We compare the baseline (Before TTA), entropy minimization-based TTA (TENT), gaussian-like soft-labeling (Soft Pseudo-Label), the proposed hard-labeling strategy (Ours), and the Oracle performance using ground-truth labels as a reference. Notably, for the *Signal Replication* fault, the values in parentheses indicate the performance achieved by applying diagnosis-guided intervention (explicit zero-masking), effectively converting the complex fault into a handleable *No Sound* case. The Overall Accuracy is reported as mean ± standard deviation across all fault types. The best performance for each metric is highlighted in bold.

SSL Model	Fault Type	Before TTA		TENT [14]		Soft Pseudo-Label		Hard Pseudo-Label (Ours)		Oracle	
		MAE (°) ↓	Acc (%) ↑	MAE (°) ↓	Acc (%) ↑	MAE (°) ↓	Acc (%) ↑	MAE (°) ↓	Acc (%) ↑	MAE (°) ↓	Acc (%) ↑
MLP-GCC [1]	No Fault (Clean)	9.46	82.24	-	-	-	-	-	-	-	-
	No Sound	45.25	25.52	22.09	45.27	19.43	47.04	**18.36**	**48.88**	18.19	58.82
	Attenuation	**9.46**	**82.25**	10.28	75.67	10.24	75.49	9.85	77.20	9.68	80.12
	Impulsive Noise	14.69	70.85	14.63	65.67	14.27	65.72	**13.77**	**67.32**	13.69	71.20
	Continuous Noise	12.14	75.67	12.64	69.47	12.45	69.47	**11.92**	**71.40**	12.03	74.80
	Signal Replication	68.73	4.66	34.11	35.55	**25.93**	**41.42**	26.05 (**18.36**)	41.20 (**48.88**)	22.73	55.76
	Average (Faults)	30.05 ± 26.02	51.79 ± 34.54	18.75 ± 9.66	58.33 ± 17.09	16.46 ± 6.29	59.83 ± 14.79	**15.99 ±6.44 (14.45 ±3.83)**	**61.20 ±15.41 (62.74 ±13.13)**	15.26 ± 5.21	68.14 ± 10.46
GCC-PHAT -SM [2]	No Fault (Clean)	7.38	83.37	-	-	-	-	-	-	-	-
	No Sound	38.46	30.01	16.92	49.86	15.29	51.51	**14.46**	**53.05**	14.12	62.45
	Attenuation	8.28	**82.23**	8.16	78.70	8.05	78.66	**7.84**	80.56	7.62	82.10
	Impulsive Noise	12.78	**73.68**	12.92	68.86	12.20	69.68	**11.59**	71.95	11.30	74.97
	Continuous Noise	10.18	**77.20**	10.31	71.76	9.68	72.96	**9.61**	74.68	9.35	77.78
	Signal Replication	59.24	6.08	30.43	35.57	22.02	42.92	**21.22** (**14.46**)	**44.07** (**53.05**)	18.93	56.61
	Average (Faults)	25.79 ± 22.35	53.84 ± 33.89	15.75 ± 8.83	60.95 ± 17.76	13.45 ± 5.52	63.15 ± 15.20	**12.94 ±5.24 (11.59 ±2.93)**	**64.86 ±15.53 (66.66 ±12.81)**	12.26 ± 4.44	70.78 ± 10.78
MPT-HiFTA [3]	No Fault (Clean)	2.65	95.29	-	-	-	-	-	-	-	-
	No Sound	11.85	77.31	5.30	85.14	4.69	85.75	**4.28**	**86.23**	3.21	91.96
	Attenuation	11.65	77.56	5.22	84.97	4.66	85.38	**4.26**	**85.80**	3.19	92.03
	Impulsive Noise	3.30	92.33	3.27	92.31	3.20	92.35	**2.99**	**92.85**	2.51	95.27
	Continuous Noise	3.53	88.91	3.50	89.28	3.42	89.30	**3.27**	**89.81**	2.48	94.94
	Signal Replication	28.83	43.34	12.83	73.37	10.00	76.77	**9.60** (**4.28**)	**77.06** (**86.23**)	3.65	91.23
	Average (Faults)	11.83 ± 10.38	75.89 ± 19.39	6.02 ± 3.92	85.01 ± 7.19	5.19 ± 2.77	85.91 ± 5.85	**4.88 ±2.70 (3.70 ±0.63)**	**86.35 ±5.93 (88.67 ±3.07)**	3.01 ± 0.50	93.09 ± 1.87

scenarios, the method yields substantial performance gains for all evaluated models. For instance, in the *No Sound* case, MLP-GCC achieves a dramatic boost, reducing MAE by 26.89° (from 45.25° to 18.36°). Similarly, MPT-HiFTA effectively recovers from *Attenuation*, improving accuracy by 8.24% points (from 77.56% to 85.80%). Notably, in the most challenging *Signal Replication* scenario, MPT-HiFTA exhibits a remarkable accuracy surge of 33.72% points (from 43.34% to 77.06%), while GCC-PHAT-SM significantly reduces its error from 59.24° to 21.22°.

Beyond structural repairs, the method consistently refines localization precision in noisy conditions; for example, MPT-HiFTA reduces MAE in *Impulsive Noise* from 3.30° to 2.99°. These comprehensive improvements culminate in a robust aggregate performance despite the inherent challenges of the continuous TTA task. Consequently, the average MAE across all faults decreases significantly for all models, closely approaching their respective Oracle benchmarks: MLP-GCC improves from 30.05° to 15.99° (Oracle: 15.26°), GCC-PHAT-SM from 24.10° to 12.94° (Oracle: 12.26°), and MPT-HiFTA from 11.83° to 4.88° (Oracle: 3.01°). Notably, the standard deviation of the average MAE also decreases consistently with improved performance, from ±10.38° to ±2.70° for MPT-HiFTA, indicating that the proposed method not only improves mean localization accuracy but also yields more consistent performance across diverse fault types. This uniform convergence validates the robustness of our self-training objective, demonstrating its capability to facilitate successful adaptation across varied model capacities.

Another critical observation is the correlation between model capacity and adaptation stability. While the high-capacity MPT-HiFTA model exhibits consistent improvements across all evaluated fault types, the lighter models (MLP-GCC and GCC-PHAT-SM) occasionally experience minor fluctuations or slight performance drops. This instability stems from the inherent challenge of continuous TTA rather than the quality of supervision signals alone. Notably, we observed similar fluctuations in lighter models even under an Oracle setting (where ground-truth labels are used for adaptation). While we empirically verified that these models eventually converge with sufficient updates, this phenomenon suggests that the frame-by-frame optimization process is

intrinsically volatile for models with limited capacity. Lighter models, due to their restricted parameter space, are more sensitive to these instantaneous gradient updates, whereas MPT-HiFTA leverages its superior representational power to buffer against such volatility, ensuring a stable adaptation trajectory.

c: COMPARISON WITH ENTROPY MINIMIZATION (TENT)

We also evaluate **TENT** [14], a widely adopted test-time adaptation method that minimizes prediction entropy. By forcing the DOA likelihood of each angular bin towards the extremes—suppressing low-probability bins while amplifying the peak probability—this objective effectively sharpens the output probability distribution. As shown in Table 3, TENT generally improves performance across all models, demonstrating that this sharpening effect helps recover spatial cues lost due to microphone faults. For instance, in the MPT-HiFTA model, TENT reduces the MAE under the *No Sound* condition by approximately 55% (from 11.85° to 5.30°).

However, despite these improvements, TENT consistently underperforms compared to our proposed Hard Pseudo-Labeling strategies. While TENT achieves an average MAE of 6.02° for MPT-HiFTA across all fault types, our Hard Pseudo-Labeling strategy attains a significantly lower error of 4.88°. This performance gap is particularly evident in complex faults like *Signal Replication*. Although TENT achieves substantial recovery in this scenario (reducing MAE from 28.83° to 12.83°), it still remains inferior to our method, which reaches 9.60°. We attribute this to the inherent limitation of indiscriminate entropy minimization: TENT adapts on all samples without a mechanism to verify prediction reliability. In contrast, our method explicitly filters out ambiguous predictions via dynamic thresholding, ensuring that the model selectively learns from high-confidence supervision, thereby achieving more precise adaptation than entropy-based approaches.

d: COMPARISON WITH SOFT PSEUDO-LABELING

To validate the effectiveness of our hard-labeling strategy, we implemented a **Soft Pseudo-Labeling** baseline for comparison. In our proposed hard-labeling approach (defined in Eq. 8), a pseudo-label is generated by assigning a probability of 1 solely to the confidence peak index k^* (i.e., a one-hot vector). In contrast, the soft baseline replaces this strict assignment with a Gaussian distribution centered at k^*, utilizing the same spatial encoding function used during training (Eq. 2). Crucially, to ensure a fair comparison, the *silence thresholding* mechanism remains identical; soft labels are generated only when the model's maximum confidence exceeds τ_{sil}, and uncertainty masking is applied to ignore low-probability bins.

Despite utilizing this refined probabilistic representation, the soft-labeling approach consistently underperforms compared to the hard-labeling strategy. We attribute this to the risk of distributing probability mass to neighboring angles under severe domain shifts. While hard labels force a sharp, decisive update on the single most confident direction, soft labels may inadvertently reinforce ambiguity or local noise surrounding the peak, thereby hindering the rapid adaptation required for fault recovery.

e: VALIDATING THE NECESSITY OF DIAGNOSIS-GUIDED ADAPTATION

The *Signal Replication* scenario serves as a critical stress test to verify the necessity of our proposed diagnosis-guided framework. As observed in Table 3, applying TTA blindly yields notable improvement for the high-capacity MPT-HiFTA model (reducing MAE from 28.83° to 9.60°). However, this recovered performance remains significantly inferior compared to other fault types, which typically converge to an MAE of approximately 3°–4°. This discrepancy arises because the replicated signals generate deceptive yet plausible spatial features; a blind adaptation algorithm struggles to fully disentangle these artifacts from the real source, resulting in suboptimal convergence.

This limitation underscores why a unified *Diagnosis + TTA* pipeline, described in Sec. IV-B4, is essential. By leveraging the precise fault classification from our diagnosis module (Table 2), our system triggers a targeted intervention, particularly for challenging faults like *Signal Replication*. Specifically, by explicitly zero-masking the corrupted channel, this intervention effectively converts the complex structural failure—characterized by conflicting spatial cues—into a geometrically tractable *No Sound* condition (**Diagnosis-Guided Intervention** strategy, as illustrated in Fig. 5). The results obtained via this diagnosis-guided strategy (reported in parentheses in Table 3) show significant performance restoration (e.g., MPT-HiFTA reaches 3.92° MAE, matching the performance of the *No Sound* case). Ultimately, these findings compellingly validate that accurate fault diagnosis is not merely an auxiliary step but a prerequisite for achieving robust and optimal adaptation against structural hardware failures.

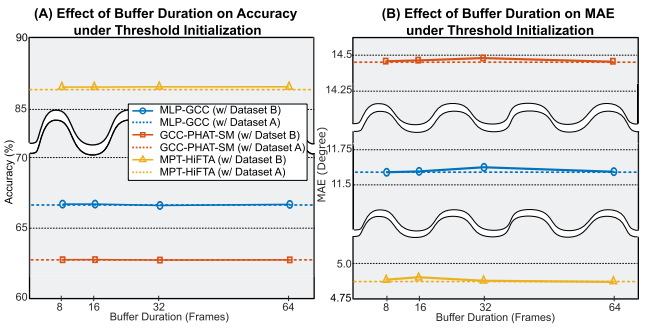


FIGURE 7. Robustness analysis of TTA performance with respect to the initialization buffer duration (N_{buffer}). We evaluate (A) Accuracy and (B) MAE across varying buffer lengths $N_{buffer} \in \{8, 16, 32, 64\}$ frames. The solid lines represent online initialization using the incoming stream (Dataset B), closely tracking the reference performance (dashed lines) derived from training set statistics (Dataset A).

3) EFFECT OF BUFFER DURATION ON THRESHOLD INITIALIZATION IN TTA

To determine the initial dynamic confidence threshold τ_0 defined in Sec. IV-B3b, we analyze the sensitivity of the TTA performance to the buffer duration used for estimating the confidence statistics. Fig. 7 illustrates the impact of varying the buffer length $N_{buffer} \in \{8, 16, 32, 64\}$ frames on both accuracy and MAE. Here, the solid lines represent the results obtained using the initialization buffer from the evaluation set (**Dataset B**), simulating an online deployment scenario where the threshold is calculated on-the-fly.

Minimal Latency with Stable Performance. The performance of all three models remains remarkably stable across different buffer durations. Specifically, increasing the buffer length from 8 to 64 frames yields negligible improvements in Accuracy and MAE. This indicates that the confidence distribution of the SSL models converges rapidly, allowing us to estimate a reliable τ_0 with a very short observation window. Consequently, we select the smallest buffer size of $N_{init} = 8$ frames for our final system. This choice significantly reduces the initialization latency, enabling the TTA module to trigger adaptation rapidly upon fault detection without accumulating a long history of data.

Robustness Across Datasets. We further validate the robustness of our threshold initialization by comparing it with an offline strategy. The dashed lines in Fig. 7 show the performance when τ_0 is pre-calculated using the training split (**Dataset A**). The results closely align with those obtained from Dataset B, showing no significant deviation. This indicates that our adaptation framework is highly resilient to variations in initialization statistics, maintaining consistent performance regardless of whether the threshold is derived from the training domain or estimated from a short window in an unseen target domain. Therefore, our method ensures reliable operation even when initialized with different statistical baselines, paving the way for the stable adaptation process described next.

Compensation via Dynamic Thresholding. To illustrate this robustness quantitatively, we observe the variation in the initialized τ_0 values across different buffer sizes. For instance, in the *Signal Replication* scenario, τ_0 varies from 0.5193 ($N_{buffer} = 8$) to 0.4040 ($N_{buffer} = 64$), fluctuating around the Dataset A baseline of 0.4471. Similarly, for *White Noise*, τ_0 ranges between 0.8041 and 0.7574, compared to 0.7650 from the training domain. Crucially, despite these variations in initialization, the final adaptation performance remains consistent. This stability is attributed to the efficacy of our Dynamic Thresholding mechanism. Even if the initial τ_0 deviates slightly due to the short observation window, the dynamic update rule allows the threshold to rapidly align with the actual confidence distribution of the test predictions during the adaptation process. A detailed analysis of this self-correcting behavior and the impact of Dynamic Thresholding is presented in the next section (Sec. VI-B4).

4) IMPACT OF DYNAMIC THRESHOLDING AND INITIALIZATION

Table 4 presents the ablation study results isolating the impact of the initialization strategy and the dynamic thresholding mechanism using the MPT-HiFTA model, chosen for its demonstrated stability. Crucially, the results presented in Table 4 were obtained without applying the proposed *Diagnosis-Guided Adaptation*; instead, we evaluated the pure blind adaptation performance across all fault types. This includes challenging scenarios like *Signal Replication*, ensuring that our parameter selection is robust even under worst-case structural anomalies.

We compare three configurations: (1) a fixed threshold ($\tau_0 = 0.8$) without updates, (2) a fixed threshold with dynamic updates, and (3) our proposed method using both adaptive initialization and dynamic updates.

Limitations of Fixed Static Thresholds. The results in the first column (Fixed $\tau_0 = 0.8$, w/o Dynamic Thres.) clearly demonstrate that a universally fixed threshold is suboptimal for diverse fault scenarios. As analyzed in the previous section, the optimal confidence distribution varies significantly by fault type; *Signal Replication* typically stabilizes around a threshold of $0.4 \sim 0.5$, whereas *Continuous (White) Noise* aligns well with a higher threshold near 0.8. Consequently, fixing τ_0 at 0.8 yields reasonable performance for *Continuous Noise* (MAE 3.28°) since the initialization matches the optimal operating point. However, for *Signal Replication*, this high static threshold is excessively strict, filtering out valid predictions and leading to a significant degradation in performance (MAE 11.53°).

Self-Correcting Capability of Dynamic Thresholding. The second column reveals the critical role of the dynamic thresholding mechanism. Even when initialized with a suboptimal value of 0.8, enabling the dynamic update rule allows the system to recover. Notably, in the *Signal Replication* scenario, the MAE improves drastically from 11.53° to 9.91°. This improvement confirms that the dynamic thresholding mechanism successfully adapts the threshold from the incorrect initial value (0.8) down to the optimal operating range ($0.4 \sim 0.5$) during the adaptation process. Conversely, for *Continuous Noise*, the performance remains stable (MAE 3.29°) because the initial value was already close to the optimal range, requiring minor adjustment.

Boosting Performance via Adaptive Initialization. Finally, the rightmost column demonstrates the additive benefit of the proposed **Adaptive Initialization**. By starting the adaptation process with a threshold estimated from a short initial buffer ($N_{init} = 8$) of the input stream (rather than an arbitrary fixed value), we observe improved performance across most metrics. This is particularly evident in challenging scenarios like *Signal Replication*, where the MAE further decreases to 9.60°. This confirms that while dynamic thresholding provides robustness against incorrect initialization, combining it with rapid on-the-fly initialization leads to a highly stable and accurate adaptation trajectory.

TABLE 4. Ablation study on confidence thresholding strategies. We evaluate the incremental contribution of the Dynamic Thresholding mechanism and the Adaptive Initialization of τ_0 using MPT-HiFTA model. The Overall Accuracy is reported as mean ± standard deviation across all fault types. The best performance for each metric is highlighted in bold.

Condition	**Fixed** $\tau_0 = 0.8$ w/o Dynamic Thres.		**Fixed** $\tau_0 = 0.8$ w/ Dynamic Thres.		**Adaptive Initialization** w/ Dynamic Thres.	
	MAE (°) ↓	Acc. (%) ↑	MAE (°) ↓	Acc. (%) ↑	MAE (°) ↓	Acc. (%) ↑
No Sound	4.79	85.75	4.45	85.94	**4.28**	**86.23**
Attenuation	4.76	85.50	4.42	85.55	**4.26**	**85.80**
Impulsive Noise	3.03	92.79	3.05	92.74	**2.99**	**92.85**
Continuous Noise	3.28	89.78	3.29	89.73	**3.27**	**89.81**
Signal Replication	11.53	74.06	9.91	76.46	**9.60**	**77.06**
Average (Faults)	5.48 ± 3.48	85.58 ±7.11	5.02 ± 2.80	86.08 ± 6.13	**4.88 ± 2.70**	**86.35 ± 5.93**

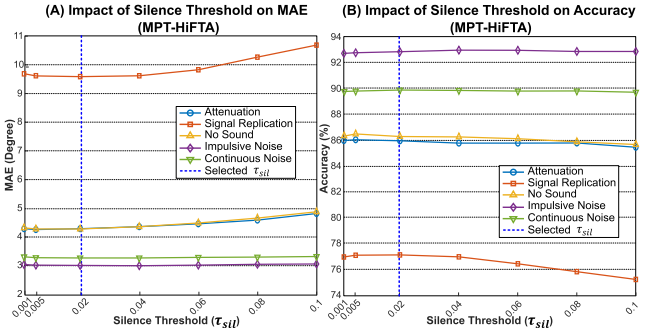


FIGURE 8. Impact of the silence threshold (τ_{sil}) on TTA performance using the MPT-HiFTA model. The results illustrate a critical trade-off: an excessively low threshold leads to silence under-detection (false negative), while an excessively high threshold causes silence over-detection (false positive).

5) IMPACT OF SILENCE THRESHOLD IN TTA

The silence threshold τ_{sil} plays a pivotal role in distinguishing active predictions from non-informative background noise. To investigate the impact of τ_{sil} on TTA performance, we conducted a sensitivity analysis using the MPT-HiFTA model, varying τ_{sil} within the range of $[0.001, 0.1]$. This analysis encompasses all fault types, including *Signal Replication*, to verify the universal applicability of the threshold.

As presented in Fig. 8, we observe a consistent behavioral trend: performance improves as the threshold increases up to a certain point, after which it begins to degrade. Based on these results, we identify $\tau_{sil} = 0.02$ as the optimal operating point. The results reveal a clear trade-off governed by the system's sensitivity to silence:

Under-detection of Silence ($\tau_{sil} < 0.02$)**:** When the threshold is set too low, the criterion for identifying silence becomes **excessively strict**. This means the system requires an extremely low confidence score to consider an input as silence. Consequently, actual silence may fail to be captured (false negative).

Over-detection of Silence ($\tau_{sil} > 0.04$)**:** Conversely, as the threshold increases, the criterion becomes overly permissive. Consequently, predictions that are not actually silence may be incorrectly selected as silence (false positive). This leads to signal loss, where potentially informative spatial cues are discarded, resulting in performance degradation.

Therefore, the choice of $\tau_{sil} = 0.02$ serves as an effective equilibrium, balancing the risks of false negative (too strict) versus false positive (too permissive). This uniform stability across diverse faults demonstrates that our proposed TTA framework is robust without requiring fault-specific hyperparameter tuning.

6) ANALYSIS OF BATCH SIZE IN TTA

To determine the optimal configuration for online TTA, we investigate the sensitivity of TTA performance to batch size variations using the MPT-HiFTA model. As observed in Fig. 9, the TTA performance remains remarkably stable across different batch sizes ($B \in \{8, 16, 32, 64\}$). Contrary to the common assumption that larger batches are essential for stabilizing batch normalization statistics, our empirical results show minimal performance fluctuation. For instance, the average MAE across all faults is 4.88° at $B = 8$ and 4.86° at $B = 32$, indicating that a small batch size provides sufficiently representative statistics for effective domain adaptation in our framework.

Based on these observations, we analyze the trade-off between adaptation accuracy and real-time responsiveness. Using a larger batch size increases the time interval between parameter updates, as the system must buffer input frames for a longer duration before performing an adaptation step. This latency can hinder the model's ability to react swiftly to sudden environmental changes or dynamic faults. Furthermore, larger buffers impose higher memory requirements, which is disadvantageous for embedded deployment. Given that $B = 8$ yields performance comparable to larger batches while maintaining a high update frequency and a low memory footprint, we select 8 as the efficient batch size for our online TTA protocol.

7) THRESHOLD SENSITIVITY ANALYSIS (τ_T)

To further investigate the robustness of the proposed TTA framework with respect to the confidence threshold, we conduct a sensitivity analysis by fixing τ_t to a specific static value across all time steps (i.e., $\tau_t = c$ for all t), rather than employing the dynamic update rule described in Sec. IV-B3. This controlled setting isolates the direct effect of the threshold magnitude on adaptation performance, independently of the dynamic update mechanism. All other

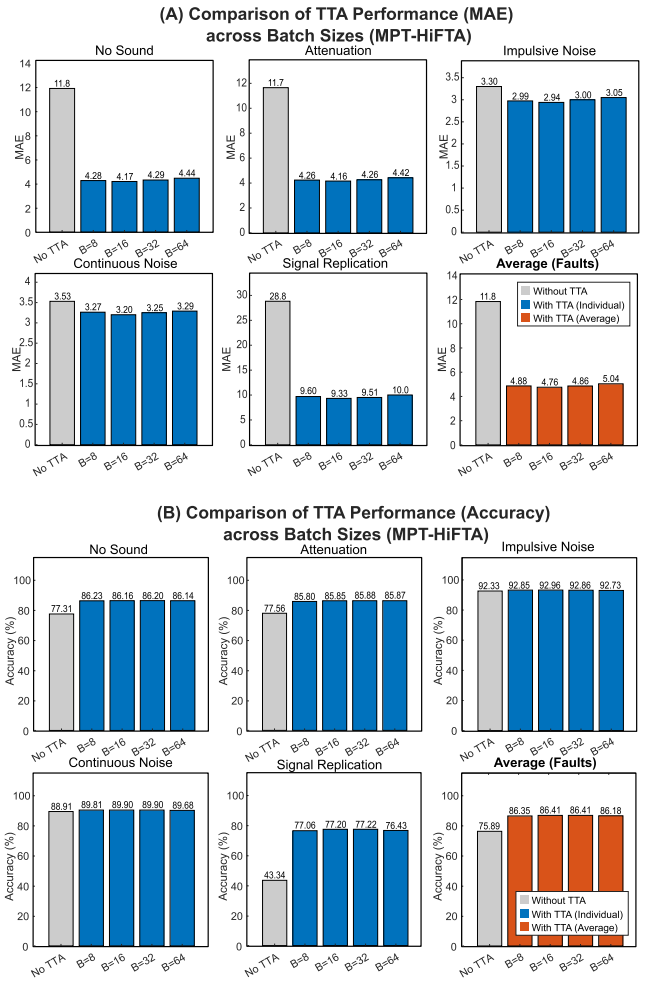


FIGURE 9. Impact of batch size on TTA performance ((A) MAE and (B) Accuracy) using the MPT-HiFTA model. The gray bars represent the baseline performance without TTA, while the colored bars indicate the adaptation performance across batch sizes ranging from 8 to 64.

hyperparameters are kept identical to those used in the preceding experiments, and we employ the MPT-HiFTA model, which demonstrated the most stable adaptation trajectories in our prior analyses.

Fig. 10 presents the results across the full range of threshold values $\tau_t \in \{0.02, 0.1, 0.2, \ldots, 0.9\}$ for all five fault types. The key observation is that the majority of fault types exhibit remarkably stable TTA performance over a broad range of static threshold values. Specifically, for *Impulsive Noise* and *Continuous Noise*—fault conditions under which MPT-HiFTA already demonstrated relatively minor performance degradation (Table 3)—the adaptation performance remains nearly constant across the entire range from 0.1 to 0.8, confirming that the SSL model retains sufficiently reliable spatial cues even under these fault conditions to support stable self-training regardless of the threshold choice. Conversely, an excessively low threshold ($\tau_t = 0.02$), which is equivalent to the silence threshold τ_{sil} and thus represents an extreme lower bound, leads to under-filtering, where noisy and unreliable predictions are admitted

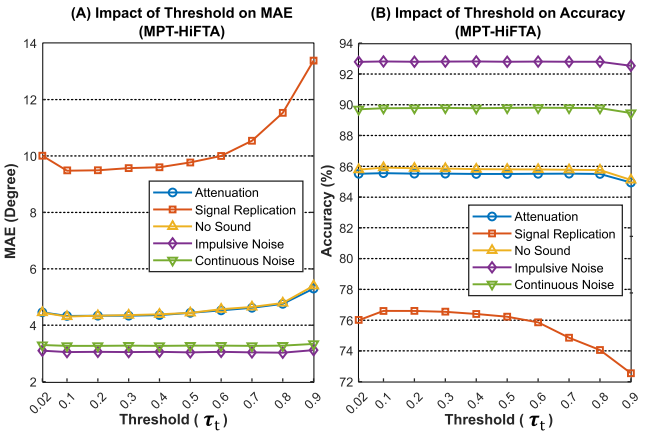


FIGURE 10. Sensitivity analysis of the confidence threshold (τ_t) on TTA performance ((A) MAE and (B) Accuracy) using the MPT-HiFTA model. To isolate the effect of the threshold magnitude independently of the dynamic update rule, τ_t is fixed to a static value across all time steps. Most fault types exhibit stable performance over a broad range, while *Signal Replication* shows stability within a narrower range due to its deceptive spatial cues.

as supervision signals, resulting in a slight but consistent performance degradation across all fault types.

TABLE 5. Performance under fault misdetection scenario with varying detection failure duration.

		True Fault: Impulsive Noise / Misdetected Fault: Signal Replication			
	Baseline	Fault Detection Failure Duration			
Metric	0s	10s	60s	300s	600s
MAE (degree)	2.99	**3.00**	3.01	3.05	3.14
ACC (%)	92.85	**92.85**	92.83	92.72	92.51

For *No Sound* and *Attenuation*, which incur moderate performance degradation before TTA, stable adaptation performance is maintained within a slightly narrower but still broad range of 0.1 to 0.7. A mild degradation is observed only at high threshold values ($\tau_t \geq 0.8$), where the overly strict filtering criterion rejects too many valid predictions, leading to insufficient supervision signals for adaptation.

In contrast, *Signal Replication* exhibits greater sensitivity to the threshold choice, yielding stable performance only within the range of approximately 0.1 to 0.5. This is consistent with our earlier observation that replicated signals generate deceptive yet low-confidence spatial cues, shifting the model's overall confidence distribution toward lower values. Consequently, a high static threshold tends to excessively filter out these already-degraded predictions, resulting in a marked performance decline for $\tau_t \geq 0.6$.

These findings provide a clear empirical basis for validating our threshold initialization and dynamic update strategies described in Sec. IV-B3. As reported in Sec. VI-B3, the adaptive initialization with $N_{\text{buffer}} = 8$ consistently estimates τ_0 within the ranges of 0.4–0.5 for *Signal Replication* and 0.75–0.80 for *Continuous Noise*, both of which fall squarely within the stable operating regions identified in this analysis.

This confirms that our data-driven initialization strategy effectively avoids suboptimal threshold values even with a minimal observation window. Furthermore, the dynamic update mechanism (Eq. 11) continuously tracks the evolving confidence distribution during adaptation, preventing the threshold from drifting into unstable regions over time and yielding performance gains over fixed threshold strategies, as demonstrated in Sec. VI-B4.

Collectively, these results demonstrate that the proposed threshold management strategy is not an arbitrary design choice but a well-grounded mechanism that ensures robust TTA performance across diverse fault conditions. Notably, a fixed initialization of $\tau_0 = 0.8$, as evaluated in Sec. VI-B4, would fall outside the stable range for Signal Replication (0.1–0.5), underscoring that adaptive initialization is not merely beneficial but essential for the most challenging fault type.

8) ANALYSIS OF TTA DEGRADATION UNDER MICROPHONE FAULT DETECTION FAILURE

To verify the robustness of the proposed framework under misdiagnosis, we analyze the system's behavior when the fault diagnosis module produces a false positive. Specifically, we simulate a scenario where the true fault is *Impulsive Noise*—the fault type that induced the least performance degradation among all evaluated conditions (Table 3)—but is misdetected as *Signal Replication*. This represents the worst-case false positive scenario, as it triggers an unnecessary channel masking intervention, silencing a channel that, despite containing impulsive noise, still retains substantial spatial information, thereby maximizing the potential loss of valid spatial cues. We assume that after the initial misdetection persists for $K = 2$ consecutive frames and triggers the erroneous intervention, the correct fault type is re-diagnosed after a varying duration of 10s, 60s, 300s, and 600s, at which point the intervention is released.

Table 5 summarizes the results. Remarkably, even under the most prolonged misdetection scenario of 600 seconds (10 minutes), the performance degradation remains marginal, with the MAE increasing by only 0.15° (from 2.99° to 3.14°) and the accuracy declining by merely 0.34% (from 92.85% to 92.51%). For shorter misdetection durations, the impact is negligible; at 10 seconds, the performance is virtually identical to the correctly diagnosed baseline.

These results highlight two important properties of the proposed framework. First, such a prolonged continuous misdetection is highly unlikely in practice, given the near-perfect diagnosis accuracy demonstrated in Table 2. Second, even in this extreme hypothetical scenario, the self-training process driven by confidence-aware pseudo-labeling partially compensates for the erroneous intervention, limiting the performance degradation to a negligible level.

It should be noted, however, that this analysis is conducted under the implicit assumption that the fault diagnosis module eventually re-diagnoses the correct fault type within a finite duration—an assumption that holds for the predefined fault types evaluated in this work, where the fault signatures are sufficiently distinct to enable eventual correct classification. For out-of-distribution fault types whose signatures do not resemble any predefined category, this self-correction mechanism may not activate, and prolonged misdetection beyond the evaluated range could occur. That said, we acknowledge that the current diagnosis module is trained on a predefined set of fault types, and its behavior under unexpected or out-of-distribution hardware failures remains an open challenge. We identify the development of more probabilistic and reversible intervention mechanisms for such unseen fault conditions as an important direction for future work (Sec. VII).

C. SIMULTANEOUS DUAL MICROPHONE FAILURE SCENARIO

1) DETECTION OF SIMULTANEOUS DUAL MICROPHONE FAILURE

While the primary experiments in this work are conducted under the single-fault assumption (Sec. III-B), real-world deployments may encounter simultaneous failures of multiple microphones. To investigate the generalizability of the proposed framework to such scenarios, we extend our evaluation to a dual microphone failure setting. Following the dataset construction protocol described in Sec. V-A2, we generate Dataset A (Dual Fault) and Dataset B (Dual Fault). For Dataset A (Dual Fault), two microphone channels and their corresponding fault types are randomly selected for each audio sample, with the same fault type applied to both channels simultaneously—reflecting the practical observation that simultaneous failures of different types across multiple microphones are rare, and concurrent faults typically share the same failure mode. For Dataset B (Dual Fault), we simulate a realistic deployment scenario where microphones 3 and 4 simultaneously experience the same fault type, extending the single-fault assumption of the original Dataset B (where only microphone 4 was faulty).

TABLE 6. Dual-fault diagnosis accuracy on Dataset A (Dual Fault) and Dataset B (Dual Fault). The results are consistent with the single-fault diagnosis results in Table 1 and Table 2, confirming that the proposed Lightweight MLP model generalizes effectively to simultaneous dual microphone failure scenarios. The Overall Accuracy is reported as mean ± standard deviation across all fault types. The best performance for each metric is highlighted in bold.

Fault Type	Dual-Fault Diagnosis Accuracy (%)			
	Dataset A (Dual Fault)		Dataset B (Dual Fault)	
	Frame-level ($K = 1$)	Queue-based ($K = 2$)	Frame-level ($K = 1$)	Queue-based ($K = 2$)
No Sound	99.99	**100.00**	99.98	**100.00**
Attenuation	99.99	**100.00**	99.95	**100.00**
Impulsive Noise	99.99	**100.00**	99.79	**100.00**
Continuous Noise	99.99	**100.00**	99.98	**100.00**
Signal Replication	99.94	**100.00**	99.77	**100.00**
Average	99.98 ± 0.02	**100.00 ±0.00**	99.89 ± 0.11	**100.00 ±0.00**

The fault diagnosis model is retrained on the training split of Dataset A (Dual Fault) following the same protocol as in Sec. V-B. Table 6 summarizes the dual-fault diagnosis accuracy on both the test split of Dataset A (Dual Fault) and Dataset B (Dual Fault). The results demonstrate near-perfect accuracy across all fault types, consistent with the single-fault results reported in Table 1 and Table 2. This confirms that the proposed Lightweight MLP model generalizes effectively to simultaneous dual microphone failure scenarios without any degradation in detection reliability.

2) TTA PERFORMANCE UNDER SIMULTANEOUS DUAL MICROPHONE FAILURE

Table 7 presents the TTA performance under the dual microphone failure scenario using the MPT-HiFTA model. As expected from the geometric constraints discussed in Sec. III-B, the Before TTA performance degrades significantly compared to the single-fault scenario (Table 3), as the simultaneous failure of two microphones reduces the available spatial information below the minimum required for reliable 2D DOA estimation. The degree of degradation is closely correlated with the amount of spatial information lost: fault types that completely destroy spatial cues (No Sound and Signal Replication) exhibit the most severe degradation (MAE of 67.58° and 50.83°, respectively), while fault types that partially preserve spatial information (Impulsive Noise and Continuous Noise) show relatively modest degradation (MAE of 5.59° and 5.27°, respectively).

TABLE 7. TTA performance under simultaneous dual microphone failure (microphones 3 and 4) using the MPT-HiFTA model. While fault types that preserve spatial information (Impulsive Noise and Continuous Noise) show recovery comparable to the single-fault scenario, fault types that completely destroy spatial cues (No Sound and Signal Replication) exhibit more limited recovery due to insufficient spatial information for reliable adaptation. The Overall Accuracy is reported as mean ± standard deviation across all fault types. The best performance for each metric is highlighted in bold.

Fault Type	Before TTA		After TTA	
	MAE (degree)	ACC (%)	MAE (degree)	ACC (%)
No Sound	67.58	12.48	**49.55**	**25.87**
Attenuation	61.39	16.74	**36.09**	**36.43**
Impulsive Noise	5.59	85.58	**4.74**	**86.53**
Continuous Noise	5.27	**79.22**	**4.70**	**79.22**
Signal Replication	50.83	14.38	**41.65**	**14.58**
Average	38.13 ± 30.45	41.68 ± 37.27	**27.35 ±21.20**	**48.53 ±32.40**

Despite these challenges, the proposed TTA framework consistently improves performance across all fault types after adaptation. For fault types that retain substantial spatial information (Impulsive Noise and Continuous Noise), the recovered performance remains comparable to the single-fault scenario, confirming the effectiveness of the self-training process even under dual failure conditions. For Attenuation, where the faulty microphones still retain some residual spatial information, TTA achieves a substantial recovery with an MAE reduction of approximately 25° (from 61.39° to 36.09°).

For No Sound and Signal Replication under dual failure conditions, however, the recovered MAE of 49.55° and 41.65° remain substantially far from the No Fault baseline (2.65°). This result is consistent with the geometric constraint discussed in Sec. III-B ($M \geq 3$): when two microphones simultaneously lose all spatial information in a 4-channel array, the remaining two operational microphones are fundamentally insufficient to resolve unambiguous 2D DOA, regardless of the adaptation strategy employed. These results therefore confirm that the practical boundary of the proposed framework—and indeed of any SSL system under this array configuration—lies not in the adaptation algorithm itself, but in the irreducible geometric constraint imposed by the hardware configuration.

D. REAL-ENVIRONMENT EVALUATION OF TTA UNDER MICROPHONE FAULT

To validate the generalizability of the proposed framework beyond the SSLR dataset and to assess its robustness against authentic hardware failures, we conduct a real-environment evaluation using a 4-channel microphone array. To physically simulate a realistic fault condition, we deliberately unplug the connector of one microphone, triggering a *No Sound* fault (Fig. 12) by disconnecting microphone channel 4. This fault type was selected as it is the most straightforward to physically reproduce in a controlled manner, whereas other fault types (e.g., Signal Replication or Continuous Noise) are difficult to induce reliably in a real hardware setting. Audio is recorded in a meeting room environment (Fig. 11) with a 4-channel microphone array placed at the center, and a loudspeaker serving as the sound source is positioned at 8 candidate locations spaced 45° apart at a distance of 1 m. Speech from the LibriSpeech test set [46] is played at each location for 1 minute per direction. To provide a reference baseline, we also record audio with all microphone connectors intact (No Fault condition).

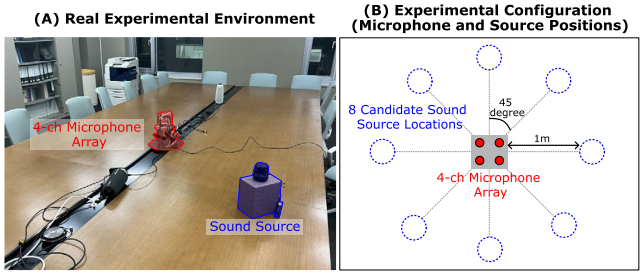


FIGURE 11. Real-environment experimental setup for validating the proposed framework. (A) The recording environment consists of a meeting room with a 4-channel microphone array placed at the center of a table.(B) The experimental configuration with 8 candidate sound source locations spaced 45° apart at a distance of 1 m from the microphone array, covering the full 360° azimuth range.

1) FINE-TUNING VIA SOUND SIMULATION

Specifically, we first generate 100,000 room impulse responses (RIRs) using Pyroomacoustics [44] with

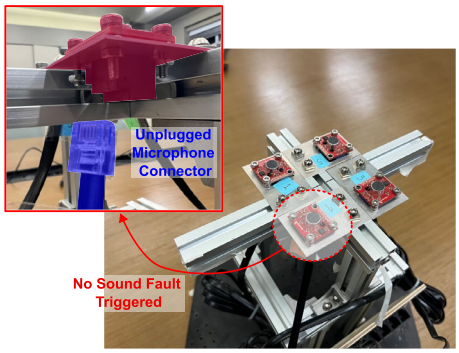


FIGURE 12. Physical reproduction of the *No Sound* fault condition. A hardware failure is deliberately induced by unplugging the connector of one microphone channel in the 4-channel array, resulting in a complete loss of signal from the affected channel.

TABLE 8. Real-environment TTA performance under an authentic No Sound fault, physically induced by unplugging a microphone connector. The fault diagnosis model achieves 100% detection accuracy, and TTA successfully recovers localization performance, confirming the generalizability of the proposed framework beyond synthetic fault conditions.

Microphone Fault Type	No Fault	No Sound Fault	
	-	Before TTA	After TTA
MAE (degree)	9.54	29.20	**16.57**
ACC (%)	82.33	61.48	**70.66**
Fault Diagnosis ACC (%)	-	100%	

randomized room geometries and source-microphone configurations that match the target array layout. We then convolve the speech recordings from the training split of the NIGENS dataset [45] with the generated RIRs to synthesize 4-channel audio with corresponding DOA labels. Specifically, each speech sample is convolved with a randomly selected RIR that is used only once and not reused across different samples, ensuring diversity in the synthesized acoustic conditions. The DOA estimation model (MPT-HiFTA) is fine-tuned on this synthetic dataset for 1 epoch using the same training protocol described in Sec. V-B, while keeping all other hyperparameters identical. For the fault diagnosis model, we apply the fault injection protocol defined in Sec. V-A1 to the same synthetic 4-channel audio to construct a fault dataset, and subsequently fine-tune the proposed Lightweight MLP for 1 epoch. Note that since the evaluation is performed on LibriSpeech recordings, there is no overlap between the NIGENS training data and the test audio, ensuring an unbiased evaluation.

2) REAL-ENVIRONMENT TTA VALIDATION

Using the fine-tuned models, we process the recorded real-world audio as follows: the fine-tuned fault diagnosis model first identifies the fault type and the affected channel, and the fine-tuned DOA estimation model then performs TTA following the protocol described in Sec. IV-B. Although our framework supports real-time streaming operation, the recorded audio is processed offline in this evaluation to ensure precise and reproducible analysis.

Table 8 summarizes the results. The fault diagnosis model achieves 100% accuracy in detecting the *No Sound* fault even at the single-frame level ($K = 1$), which is consistent with the near-perfect frame-level accuracy reported in Table 2 and confirms that the diagnosis model generalizes reliably to authentic hardware failures. Following fault detection, the proposed TTA framework successfully recovers localization performance, reducing the MAE from 29.20° to 16.57° and improving accuracy from 61.48% to 70.66%. While this recovered performance remains below the No Fault baseline (9.54° MAE, 82.33% accuracy), we attribute this gap primarily to two factors.

First, the Sim-to-Real discrepancy inherent in simulation-based fine-tuning introduces a performance ceiling, consistent with observations in [3] where similar degradation was reported when models trained on simulated data are applied to real recordings. Second, the total available data for TTA is limited to approximately 8 minutes of recordings, which is significantly less than the SSLR test set used in the main experiments. We expect that collecting more diverse real-world data and extending the adaptation duration would further close this gap. Collectively, these results validate that the proposed framework is not only effective on the SSLR benchmark but also generalizes to real-world hardware fault scenarios.

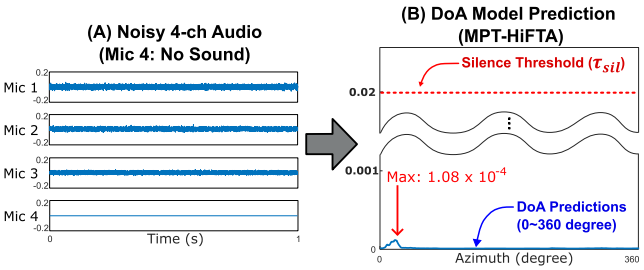


FIGURE 13. Visualization of the DOA model prediction under the *No Sound* fault condition with additional background noise (SNR: −8.34 dB). (A) The 4-channel audio input where Mic 4 is silent due to the hardware fault, and the remaining channels contain both speech and background noise. (B) The corresponding DOA likelihood predictions, where the maximum prediction value (1.08×10^{-4}) remains well below the static silence threshold ($\tau_{sil} = 0.02$), confirming that the silence threshold effectively suppresses noisy inputs even under severe background noise conditions.

3) ROBUSTNESS OF SILENCE THRESHOLD UNDER NOISY CONDITIONS

To address the concern that the static silence threshold τ_{sil} may become unreliable under varying background noise levels, we conduct an additional analysis to verify its robustness under severe noise conditions. Specifically, using the same real-environment setup described in Sec. VI-D2, we record 10 seconds of audio with the *No Sound* fault active on microphone channel 4, while simultaneously introducing additional background white noise at an SNR of −8.34 dB relative to the previously recorded speech signal. This noise level corresponds to a severely degraded acoustic environment, comparable to conditions between a noisy

café (approximately −5 dB SNR) and loud machinery noise (approximately −10 dB SNR).

The recorded noisy audio is fed into the fine-tuned DOA estimation model (MPT-HiFTA), and we examine the maximum prediction value across all intra-frame steps to verify whether the static silence threshold remains effective in suppressing non-source predictions. As illustrated in Fig. 13, the DOA likelihood predictions for a representative 1-second segment remain consistently well below the static silence threshold ($\tau_{\text{sil}} = 0.02$), with a maximum prediction value of 1.08×10^{-4}. Furthermore, this behavior is consistent across the entire 10-second recording, with the average maximum prediction over all segments measured at 1.0771×10^{-4}, confirming that the model produces uniformly low-confidence predictions throughout the noisy silence period.

These results demonstrate that the static silence threshold $\tau_{\text{sil}} = 0.02$ provides a sufficiently large margin against severe background noise, effectively preventing noisy inputs from being misclassified as valid source predictions. We attribute this robustness to the inherent noise-resilience of the pre-trained DOA estimation model, which is specifically designed and trained to handle noisy acoustic conditions. Consequently, even in the presence of severe background noise, the silence threshold reliably suppresses non-source regions, ensuring the stability of the pseudo-labeling process during TTA.

VII. CONCLUSION AND FUTURE WORKS

In this paper, we proposed a comprehensive Fault-Resilient SSL framework that ensures reliable robotic perception even under severe hardware failures. By tightly integrating a Lightweight Frequency-Time Projection Network for fault diagnosis with a Diagnosis-Guided Confidence-Aware TTA strategy, our system autonomously identifies sensor malfunctions and adapts the model at test-time. Experimental results demonstrate that the proposed method effectively mitigates performance degradation, reducing the MAE from 21.99° to 9.91° averaged across diverse models and fault conditions. Furthermore, we validate the generalizability of the proposed framework through a real-environment experiment using a physical microphone array with an authentic hardware fault, confirming that the framework is not limited to synthetic fault conditions but extends to real-world hardware failures.

Our findings highlight two critical insights for robust auditory perception. First, we demonstrated that accurate fault diagnosis is a prerequisite for effective adaptation; specifically, our diagnosis-guided intervention strategy—which masks corrupted channels based on precise fault classification—was essential for recovering from structural failures like signal replication, where unsupervised adaptation alone proved insufficient. Second, our confidence-aware hard pseudo-labeling approach significantly outperformed entropy minimization and soft-labeling baselines, demonstrating that selectively retaining reliable predictions while filtering out ambiguous ones is key to preventing error accumulation during self-training.

For future work, we plan to extend the scalability and applicability of this framework in three key directions. First, as demonstrated in Sec. VI-C, simultaneous failures of multiple microphones can violate the geometric constraint ($M \geq 3$) required for reliable 2D DOA estimation, imposing a fundamental limitation that cannot be overcome by adaptation alone. Addressing this challenge likely requires extending beyond single-modality auditory perception toward multimodal approaches—for instance, leveraging complementary visual or inertial cues to compensate for the spatial information lost due to microphone failures, thereby enabling robust localization even under severe array degradation. Second, we intend to apply our fault-resilient paradigm to other array-based tasks, such as multi-channel automatic speech recognition, to investigate how diagnosis-guided adaptation can maintain recognition performance in the presence of hardware degradation.

Third, while the proposed fault diagnosis module achieves near-perfect accuracy on the predefined set of fault types evaluated in this work, its reliability may degrade when confronted with unexpected or out-of-distribution hardware failures that were not encountered during training. This is an inherent limitation of classification-based fault diagnosis approaches, including both the conventional VAD-based method [10] and our proposed model, as classification-based formulations are fundamentally constrained to recognizing fault types seen during training and cannot naturally generalize to unseen fault patterns. In such cases, an erroneous diagnosis could trigger an inappropriate intervention, potentially compromising localization performance. Although our analysis in Sec. VI-B8 demonstrates that even prolonged misdetection results in only marginal performance degradation, addressing this limitation remains an important open challenge that goes beyond the scope of classification-based diagnosis alone. Future work should therefore explore more probabilistic [47], gradual [48], or reversible intervention [34] mechanisms that can gracefully handle unseen fault conditions without placing absolute trust in the diagnosis module's output.

REFERENCES

[1] W. He, P. Motlicek, and J.-M. Odobez, "Deep neural networks for multiple speaker detection and localization," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2018, pp. 74–79.

[2] J. Wang, X. Qian, Z. Pan, M. Zhang, and H. Li, "GCC-PHAT with speech-oriented attention for robotic sound source localization," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2021, pp. 5876–5883.

[3] I. An, G. An, T. Kim, and S.-E. Yoon, "Microphone pair training for robust sound source localization with diverse array configurations," *IEEE Robot. Autom. Lett.*, vol. 9, no. 1, pp. 319–326, 2023.

[4] I. An, Y. Kwon, and S.-E. Yoon, "Diffraction- and reflection-aware multiple sound source localization," *IEEE Trans. Robot.*, vol. 38, no. 3, pp. 1925–1944, Jun. 2022.

[5] I. An, B. Jo, Y. Kwon, J.-W. Choi, and S.-E. Yoon, "Robust sound source localization considering similarity of back-propagation signals," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2020, pp. 1574–1580.

[6] H. Liu, Z. Zhang, Y. Zhu, and S.-C. Zhu, "Self-supervised incremental learning for sound source localization in complex indoor environment," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2019, pp. 2599–2605.

[7] M. Brandstein and D. Ward, *Microphone Arrays: Signal Processing Techniques and Applications*. Cham, Switzerland: Springer, 2001.

[8] J. Benesty, J. Chen, and Y. Huang, *Microphone Array Signal Processing*. Cham, Switzerland: Springer, 2008.

[9] B. Rafaely, *Fundamentals of Spherical Array Processing*, vol. 8. Cham, Switzerland: Springer, 2015.

[10] J. Kim and B.-J. You, "Fault detection in a microphone array by intercorrelation of features in voice activity detection," *IEEE Trans. Ind. Electron.*, vol. 58, no. 6, pp. 2568–2571, Jun. 2011.

[11] M. Uricár, P. Krížek, G. Sistu, and S. Yogamani, "SoilingNet: Soiling detection on automotive surround-view cameras," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Oct. 2019, pp. 67–72.

[12] H. Ryu, M. Yoon, D. Park, and S.-E. Yoon, "Confidence-based robot navigation under sensor occlusion with deep reinforcement learning," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 8231–8237.

[13] X. Chen, A. Bettens, Z. Xie, Z. Wang, and X. Wu, "Kalman filter and neural network fusion for fault detection and recovery in satellite attitude estimation," *Acta Astronautica*, vol. 217, pp. 48–61, Apr. 2024.

[14] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," 2020, *arXiv:2006.10726*.

[15] C. Finn, S. Levine, and M. Zhang, "MEMO: Test time robustness via adaptation and augmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 38629–38642.

[16] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, and M. Tan, "Efficient test-time model adaptation without forgetting," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 16888–16905.

[17] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, and M. Tan, "Towards stable test-time adaptation in dynamic wild world," 2023, *arXiv:2302.12400*.

[18] M. Jang, S.-Y. Chung, and H. W. Chung, "Test-time adaptation via self-training with nearest neighbor information," 2022, *arXiv:2207.10792*.

[19] Q. Wang, O. Fink, L. Van Gool, and D. Dai, "Continual test-time domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 7201–7211.

[20] Y. Yu, S. Shin, S. Back, M. Ko, S. Noh, and K. Lee, "Domain-specific block selection and paired-view pseudo-labeling for online test-time adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 22723–22732.

[21] Y. Hu, C. Qiao, X. Geng, and N. Xu, "Selective label enhancement learning for test-time adaptation," in *Proc. 13th Int. Conf. Learn. Represent.*, 2025. [Online]. Available: https://openreview.net/forum?id=3Z2flzXzBY

[22] H. Park, A. Gupta, and A. Wong, "Test-time adaptation for depth completion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 20519–20529.

[23] J. Yoon, S. Kim, S. Kwak, and M. Cho, "Optical flow domain adaptation via target style transfer," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 2111–2121.

[24] Q. Lin, R. Chen, K. Gu, and A. Yao, "Semantics-aware test-time adaptation for 3D human pose estimation," 2025, *arXiv:2502.10724*.

[25] T. Lee, J. Tremblay, V. Blukis, B. Wen, B.-U. Lee, I. Shin, S. Birchfield, I. S. Kweon, and K.-J. Yoon, "TTA-COPE: Test-time adaptation for category-level object pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 21285–21295.

[26] D. Su, T. Vidal-Calleja, and J. V. Miro, "Towards real-time 3D sound sources mapping with linear microphone arrays," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2017, pp. 1662–1668.

[27] L. Fu, Y. He, J. Wang, X. Qiao, and H. Kong, "I-ASM: Iterative acoustic scene mapping for enhanced robot auditory perception in complex indoor environments," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2024, pp. 12318–12323.

[28] V. Venkatasubramanian, R. Rengaswamy, S. N. Kavuri, and K. Yin, "A review of process fault detection and diagnosis: Part III: Process history based methods," *Comput. Chem. Eng.*, vol. 27, no. 3, pp. 327–346, 2003.

[29] S. X. Ding, *Model-Based Fault Diagnosis Techniques: Design Schemes, Algorithms, and Tools*. Cham, Switzerland: Springer, 2008.

[30] M. Zhao, S. Zhong, X. Fu, B. Tang, and M. Pecht, "Deep residual shrinkage networks for fault diagnosis," *IEEE Trans. Ind. Informat.*, vol. 16, no. 7, pp. 4681–4690, Jul. 2020.

[31] T. Mezair, Y. Djenouri, A. Belhadi, G. Srivastava, and J. C.-W. Lin, "A sustainable deep learning framework for fault detection in 6G industry 4.0 heterogeneous data environments," *Comput. Commun.*, vol. 187, pp. 164–171, Apr. 2022.

[32] U. Ali, U. Ramzan, W. Ali, and K. A. Al-Jaafari, "An improved fault diagnosis strategy for induction motors using weighted probability ensemble deep learning," *IEEE Access*, vol. 13, pp. 106958–106973, 2025.

[33] H. Ha, S. Han, and J. Lee, "Fault detection on transmission lines using a microphone array and an infrared thermal imaging camera," *IEEE Trans. Instrum. Meas.*, vol. 61, no. 1, pp. 267–275, Jan. 2012.

[34] W. Zhang, H. Zhou, S. Sun, Z. Wang, J. Shi, and C. C. Loy, "Robust multi-modality multi-object tracking," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Aug. 2019, pp. 2365–2374.

[35] S. Cicero, M. Guarascio, A. Guerrieri, S. Mungari, and A. Vinci, "A deep neural framework for fault detection in IoT-based sensor networks," in *Proc. Int. Conf. Adv. Social Netw. Anal. Mining*. Cham, Switzerland: Springer, 2024, pp. 129–145.

[36] G. Zheng, Q. Zhang, and S. Li, "Failure diagnosis of antenna array using residual convolutional neural network," in *Proc. Int. Appl. Comput. Electromagn. Soc. (ACES-China) Symp.*, Jul. 2021, pp. 1–2.

[37] L. A. L. Janssen and I. L. Arteaga, "Data processing and augmentation of acoustic array signals for fault detection with machine learning," *J. Sound Vibrat.*, vol. 483, Sep. 2020, Art. no. 115483.

[38] C. Li, C. Chen, and X. Gu, "Acoustic-based rolling bearing fault diagnosis using a co-prime circular microphone array," *Sensors*, vol. 23, no. 6, p. 3050, Mar. 2023.

[39] L. Chen, Y.-S. Choy, K.-C. Tam, and C.-W. Fei, "Hybrid microphone array signal processing approach for faulty wheel identification and ground impedance estimation in wheel/rail system," *Appl. Acoust.*, vol. 172, Jan. 2021, Art. no. 107633.

[40] B. Zhou, K. Xu, Y. Quan, and M. Xing, "TransDOA: Calibrating array imperfections via transformer-based transfer learning," 2025, *arXiv:2504.13394*.

[41] C. Knapp and G. Carter, "The generalized correlation method for estimation of time delay," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-24, no. 4, pp. 320–327, Aug. 1976.

[42] S. Adavanne, A. Politis, J. Nikunen, and T. Virtanen, "Sound event localization and detection of overlapping sources using convolutional recurrent neural networks," *IEEE J. Sel. Topics Signal Process.*, vol. 13, no. 1, pp. 34–48, Mar. 2019.

[43] R. A. Marsden, M. Döbler, and B. Yang, "Introducing intermediate domains for effective self-training during test-time," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2024, pp. 1–10.

[44] R. Scheibler, E. Bezzam, and I. Dokmanic, "Pyroomacoustics: A Python package for audio room simulation and array processing algorithms," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2018, pp. 351–355.

[45] I. Trowitzsch, J. Taghia, Y. Kashef, and K. Obermayer, "The NIGENS general sound events database," 2019, *arXiv:1902.08314*.

[46] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: An ASR corpus based on public domain audio books," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2015, pp. 5206–5210.

[47] J. Ren, J. Wen, Z. Zhao, R. Yan, X. Chen, and A. K. Nandi, "Uncertainty-aware deep learning: A promising tool for trustworthy fault diagnosis," *IEEE/CAA J. Autom. Sinica*, vol. 11, no. 6, pp. 1317–1330, Jun. 2024.

[48] Y. Fu and W. Yan, "One masked model is all you need for sensor fault detection, isolation and accommodation," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2024, pp. 1–8.



INKYU AN received the B.S. degree in electronics engineering from Dongguk University, in 2016, the M.S. degree in robotics program from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea, in 2018, and the Ph.D. degree from the School of Computing, KAIST, in 2023. He was a Researcher with the Artificial Intelligence Research Laboratory, Electronics and Telecommunications Research Institute (ETRI), Daejeon. He is currently an Assistant Professor with Kookmin University. His research interests include robot audition, e.g., speech separation, sound source localization and detection, and sound-aware multimodal embodied artificial intelligence (AI).